

# Digital Design using FPGA

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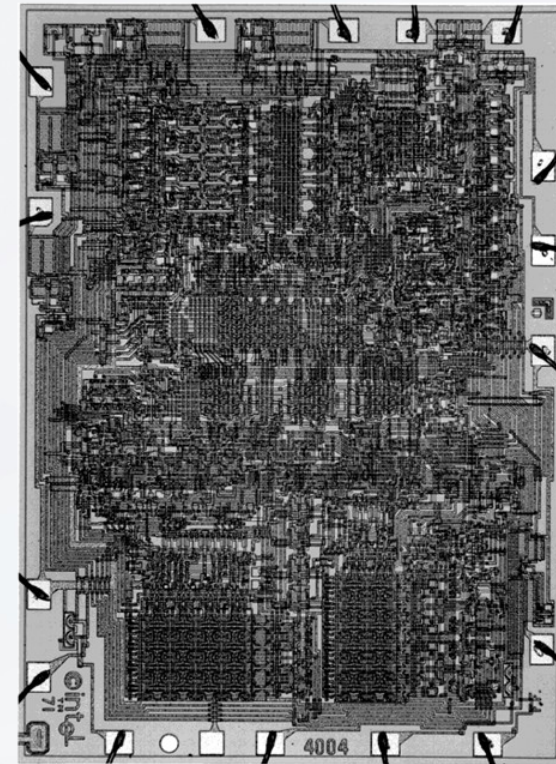


# Target Audience

Interested in learning Digital IC Design

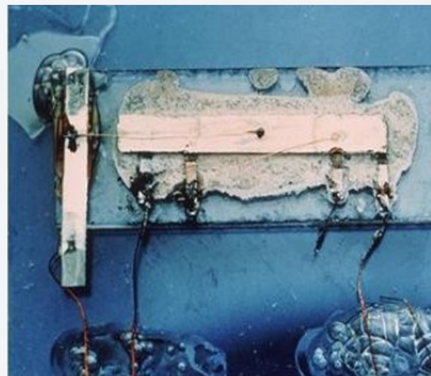
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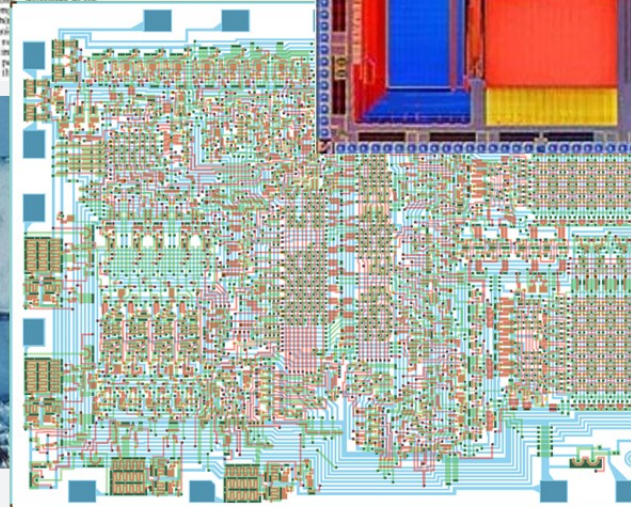




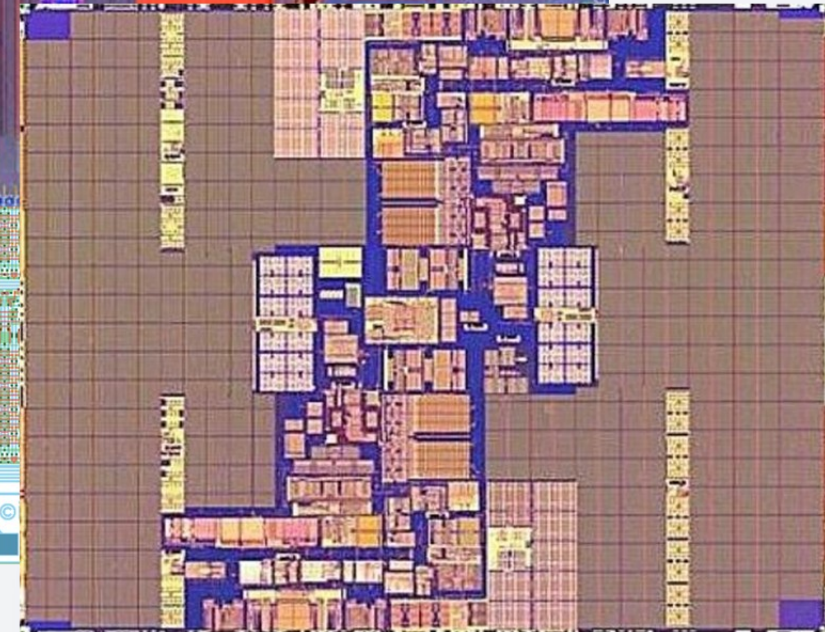
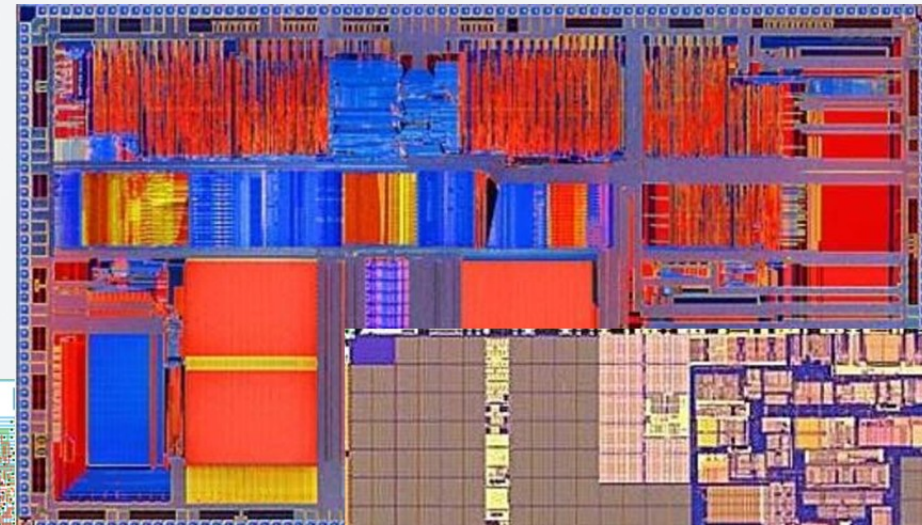
# Motivation



**First Integrated circuit, 1964**



**The Intel 4004, 1971  
2,300 Transistors**

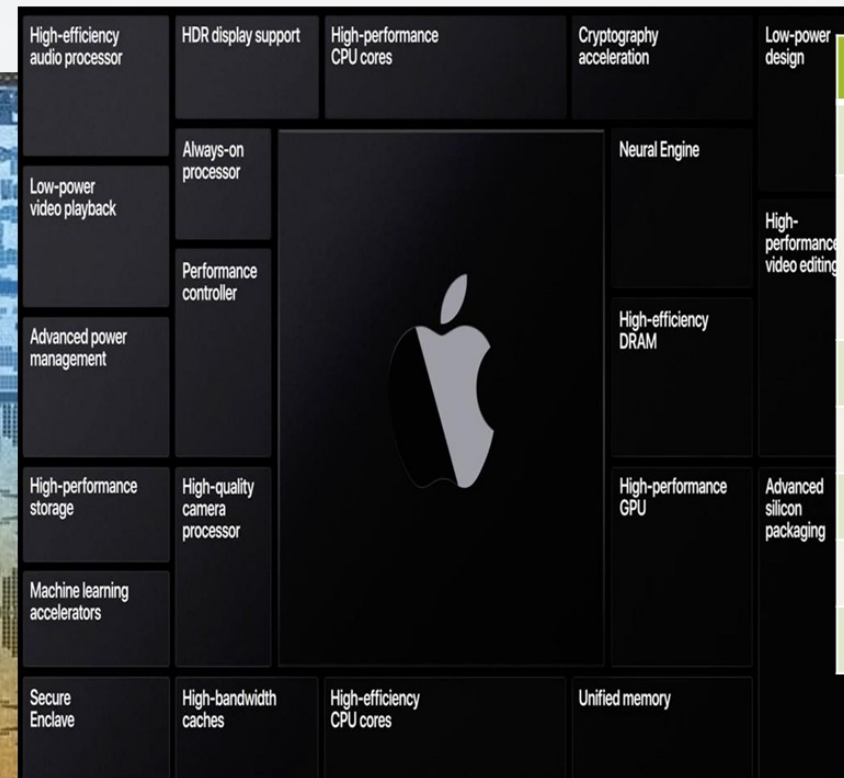
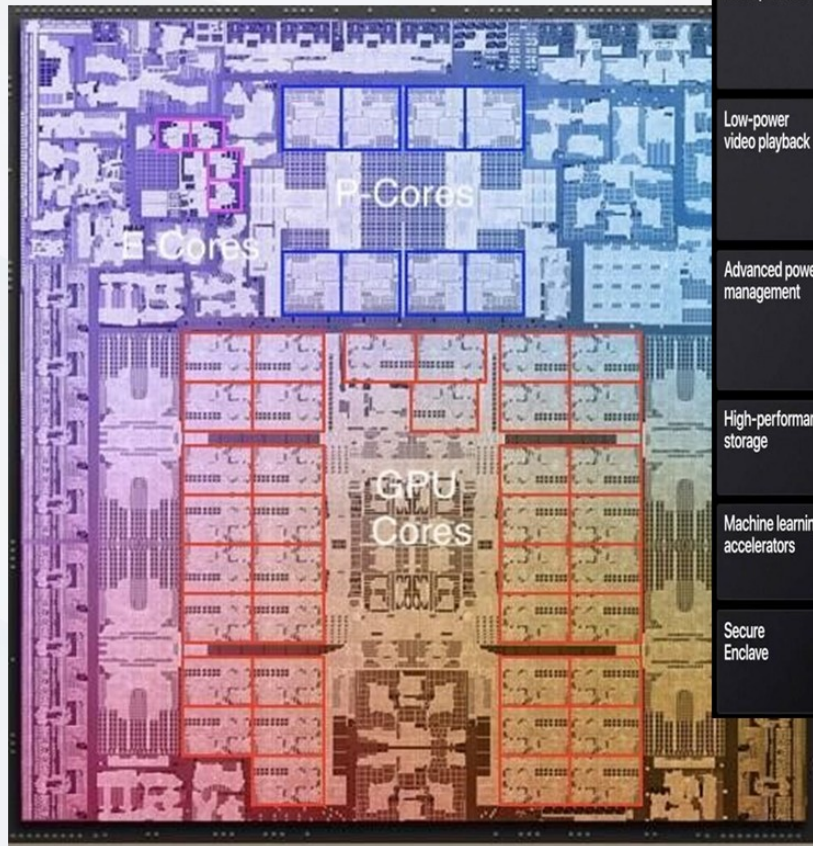


**Itanium 2  
"Montecito", 2006  
1.7B Transistors**



# Motivation -2

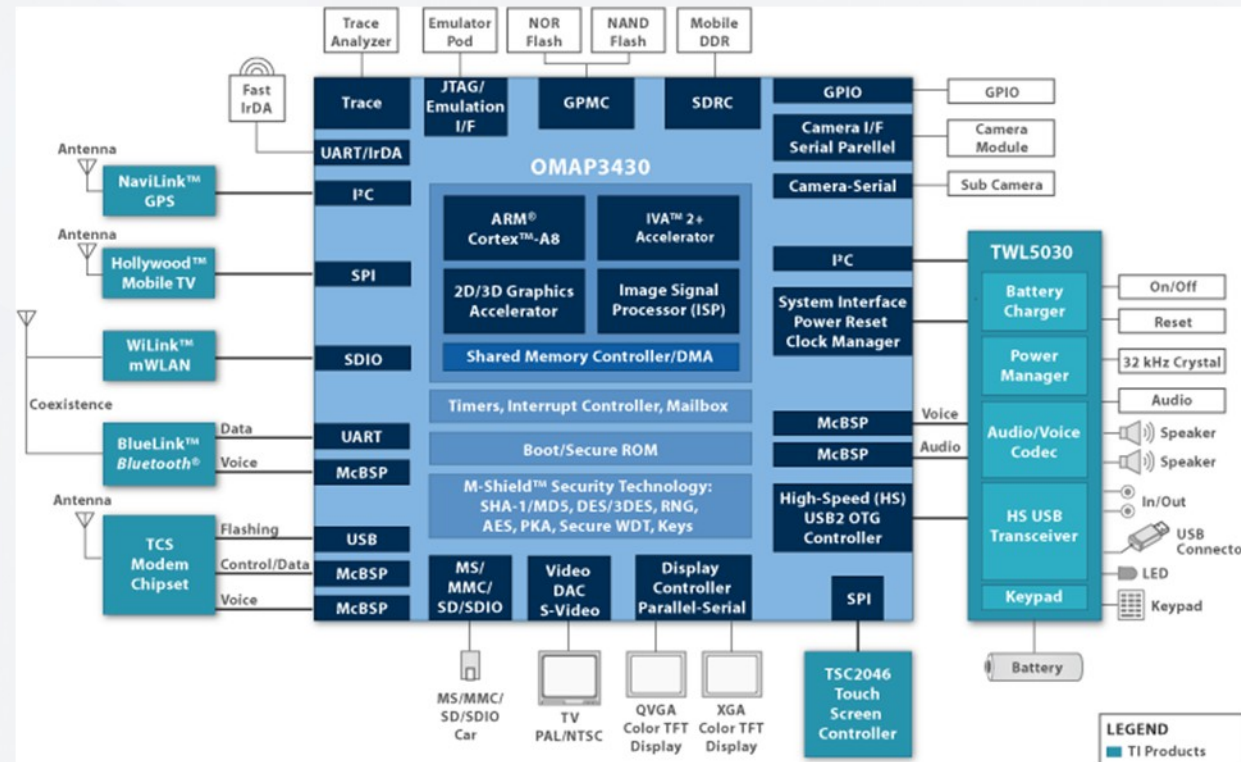
## Apple M3



| Introduced   | 2023                            |
|--------------|---------------------------------|
| Technology   | TSMC 3nm                        |
| memory       | 2MB L1<br>20MB L2\$<br>8MB L3\$ |
| Cores        | 16                              |
| Threads      | 16                              |
| Frequency    | 2.8- 4.06 GHz                   |
| Die Size     | 141.7 mm <sup>2</sup>           |
| #Transistors | 92 B                            |

# Motivation -3

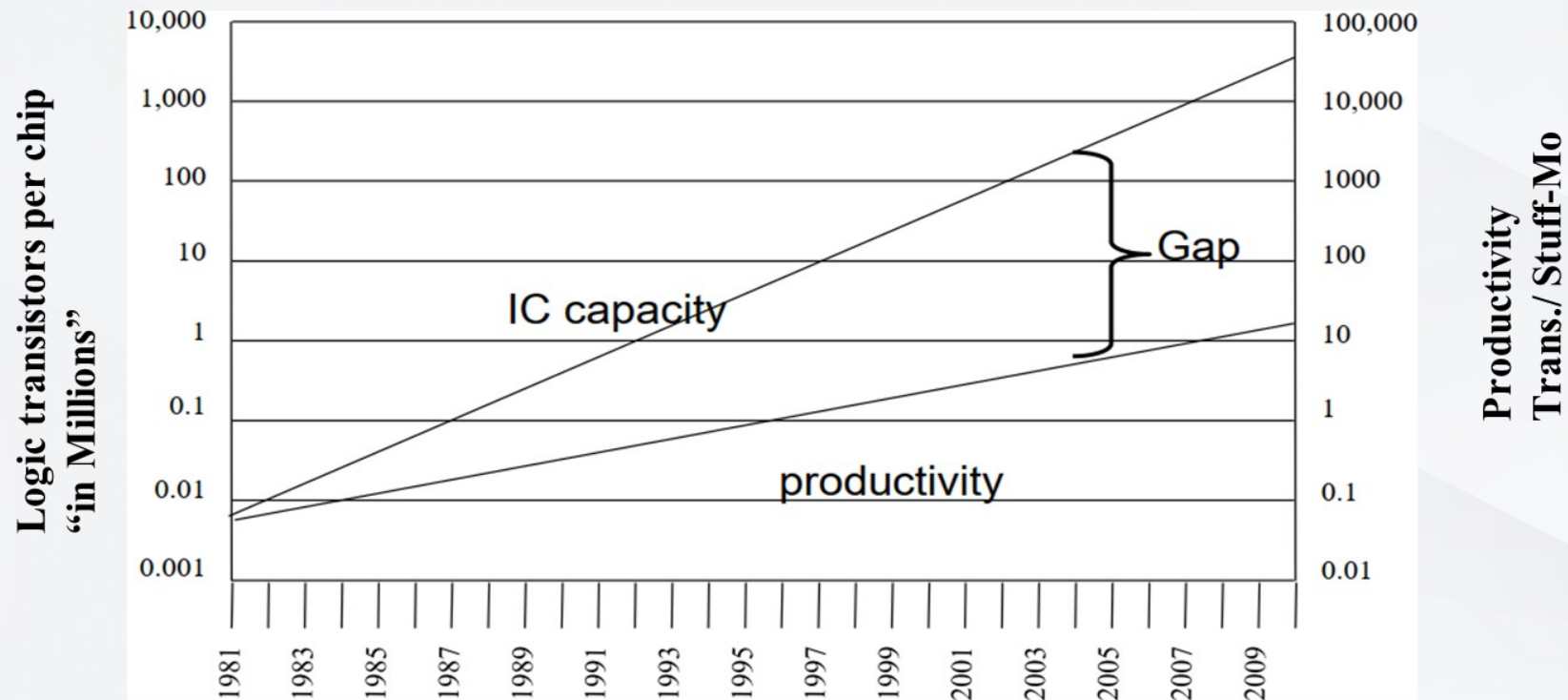
- How on earth do we design such a Monster???



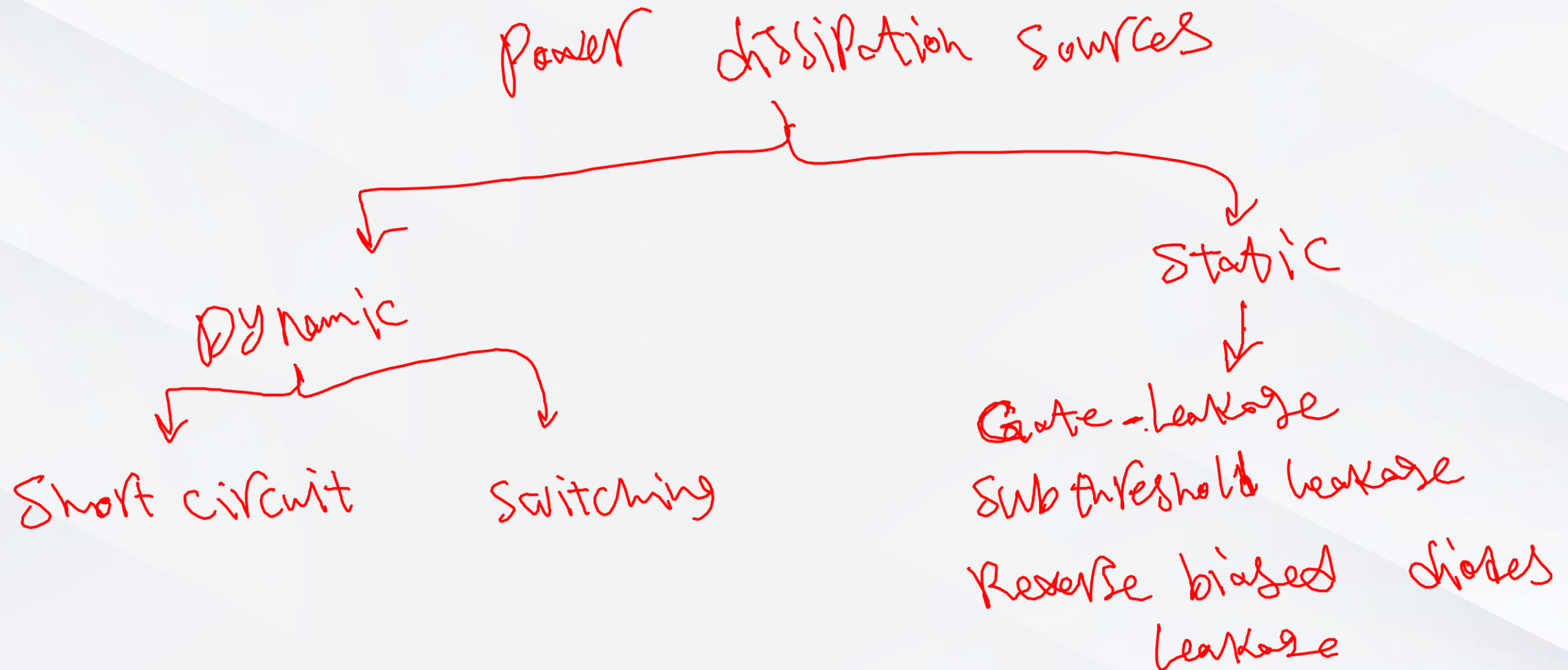


# Motivation -4

- Hummmm!, we have a problem...



“Moore’s Law of Engineers”

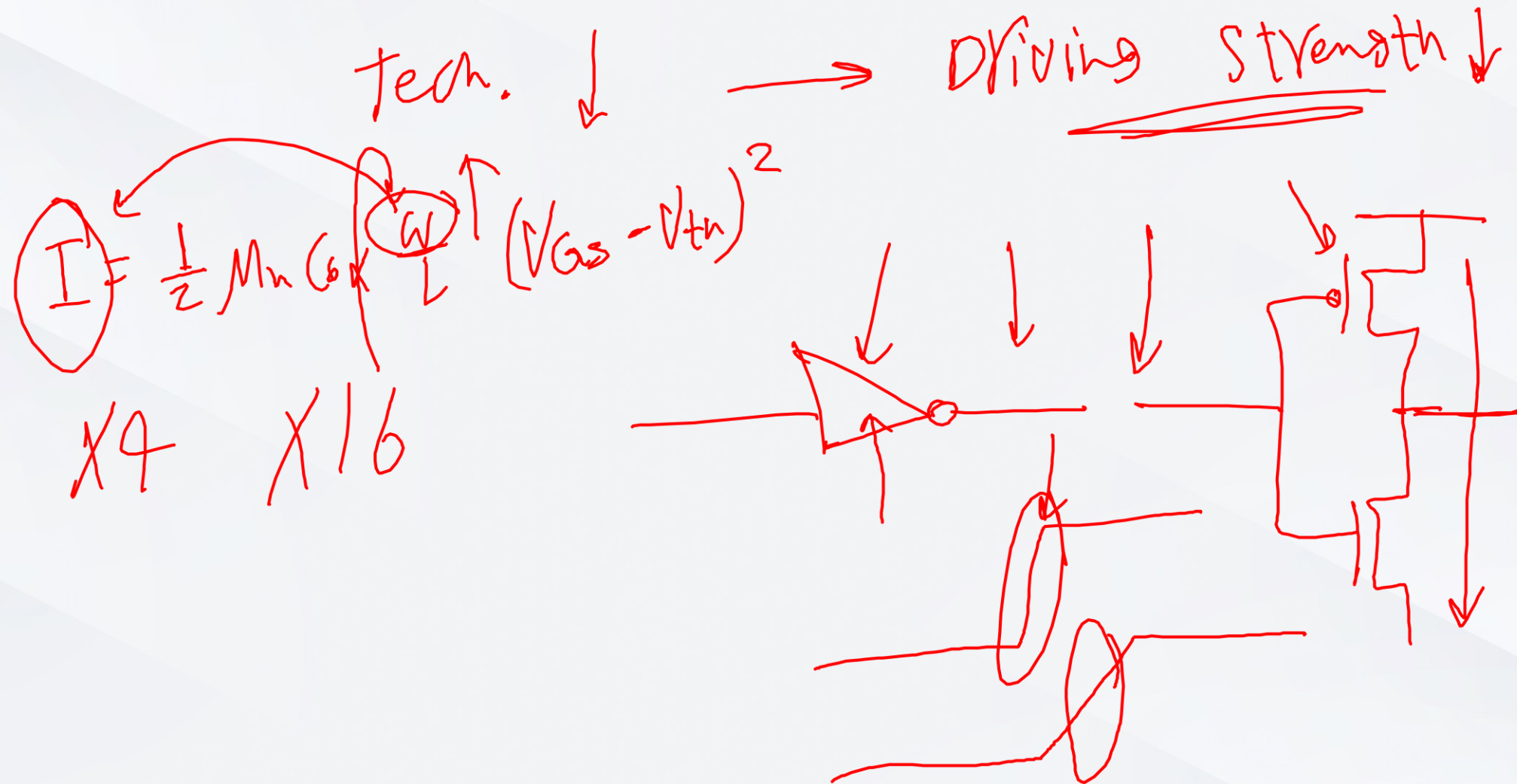


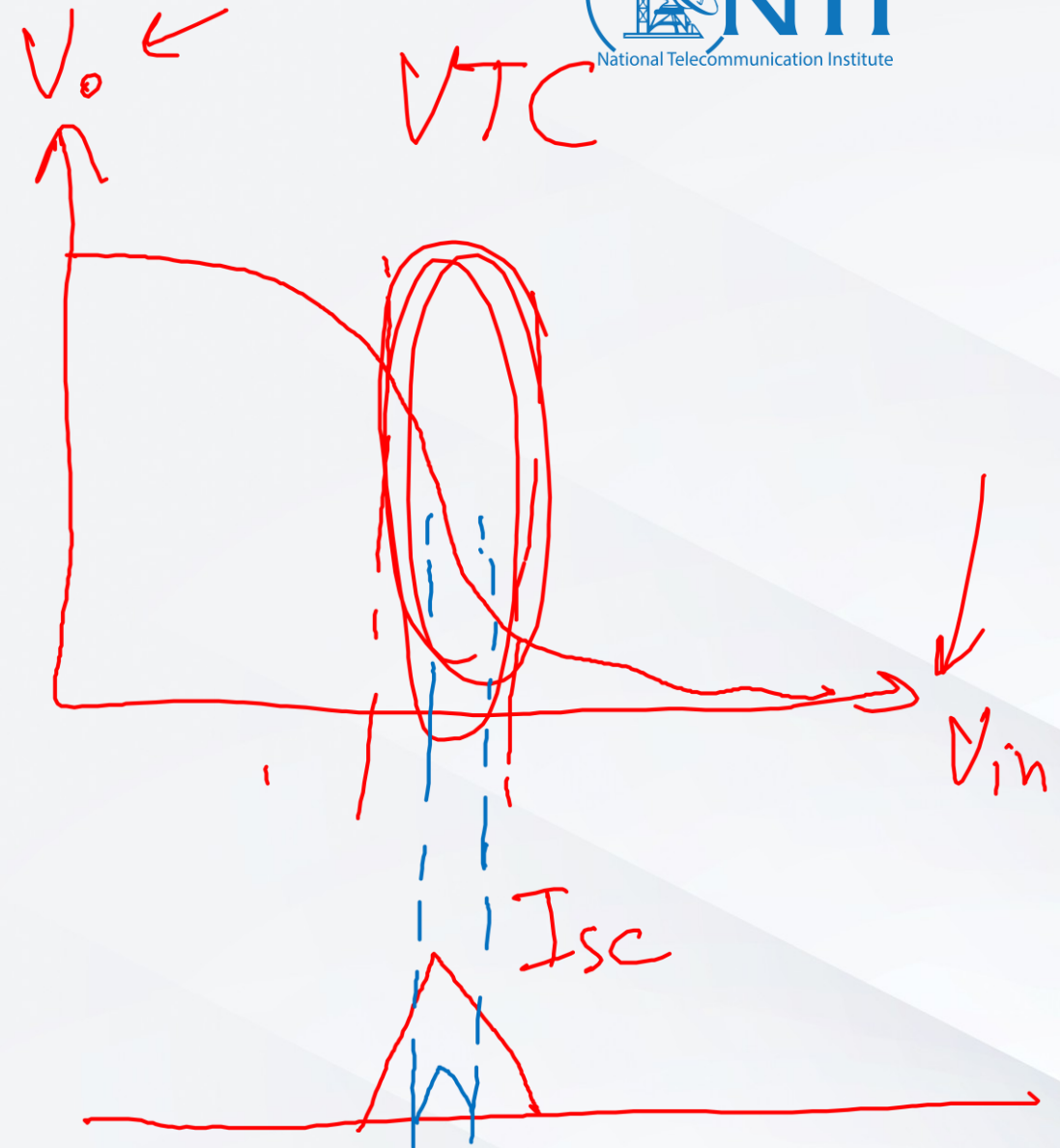
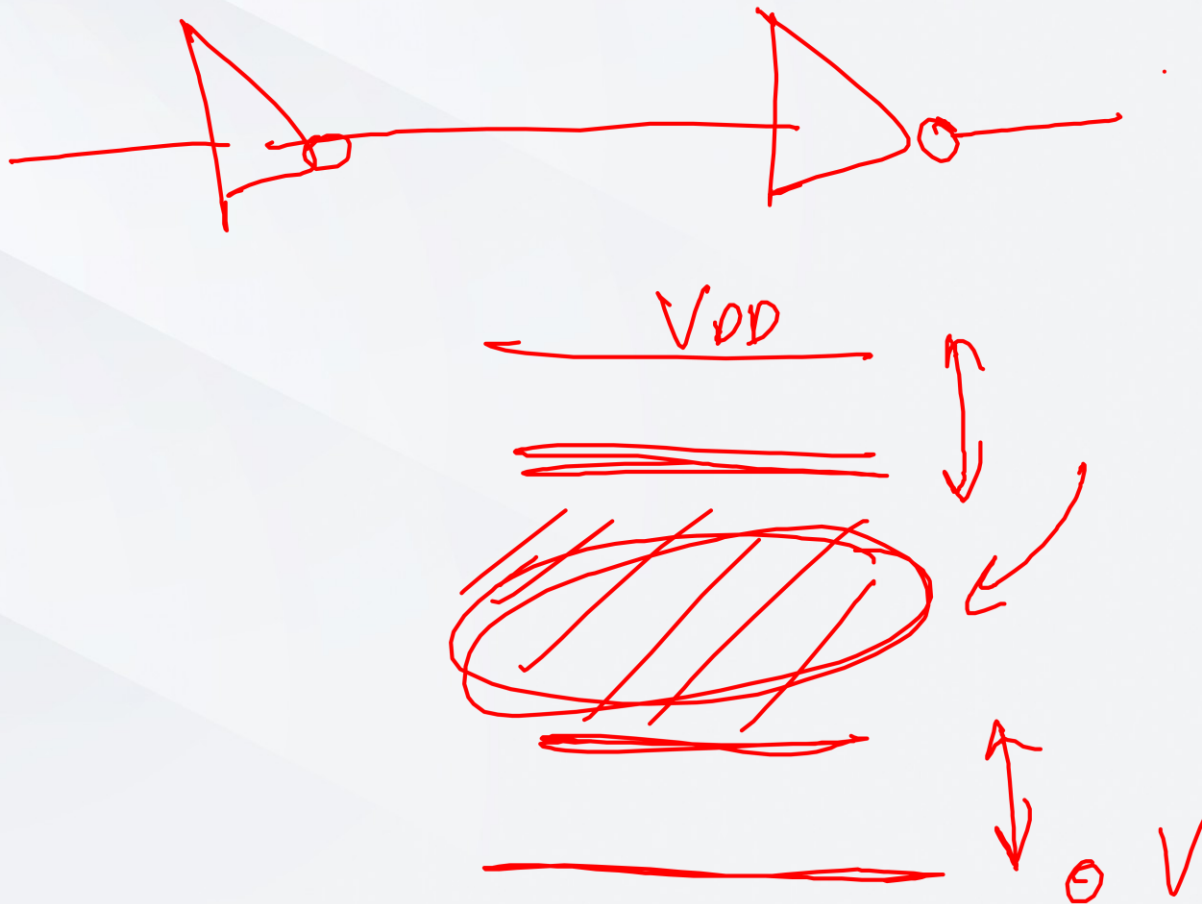


switching power

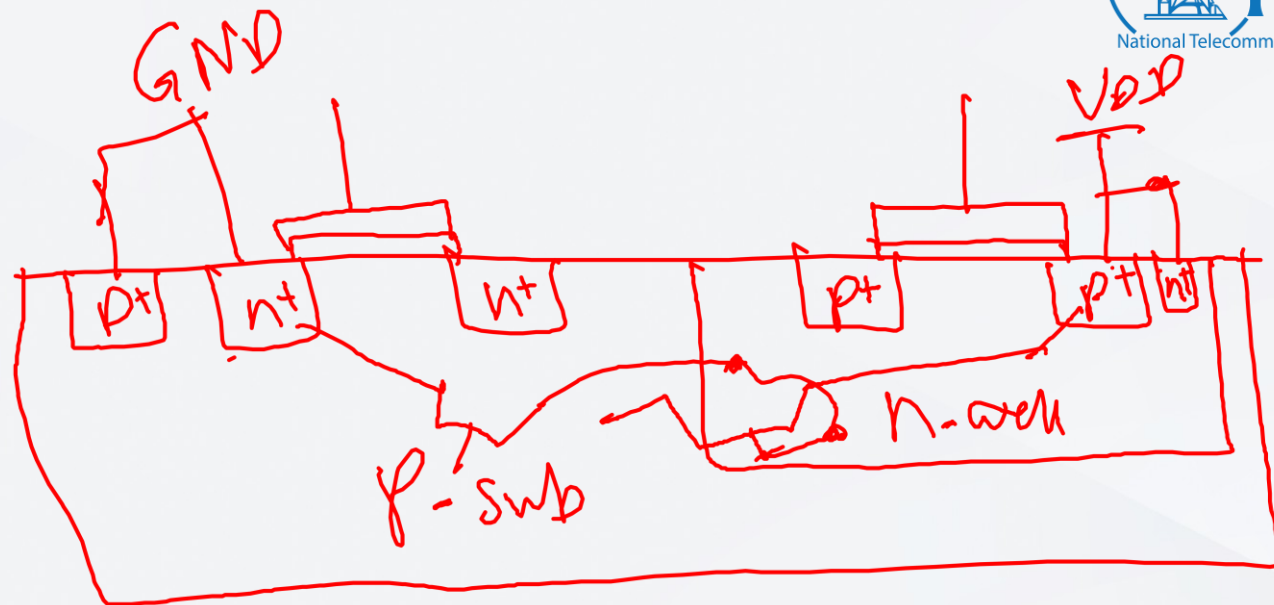
$$P_{sw} = \alpha \underbrace{V_{DD}^2} \underbrace{C_L} \underbrace{f} \rightarrow f_{CLK}$$

Glitches









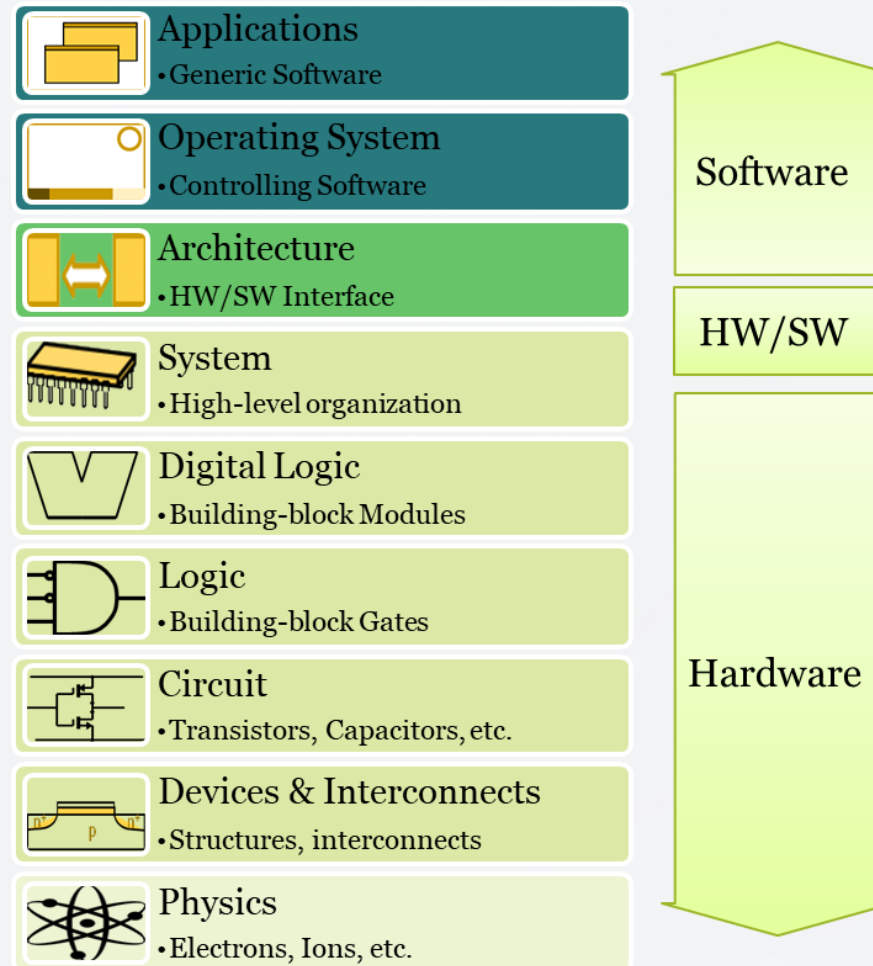
# Motivation-5

**Design  
Abstraction**

**Design  
Automation  
“EDA”**

**Design Re-use  
“IP”**

# Design Abstraction Levels

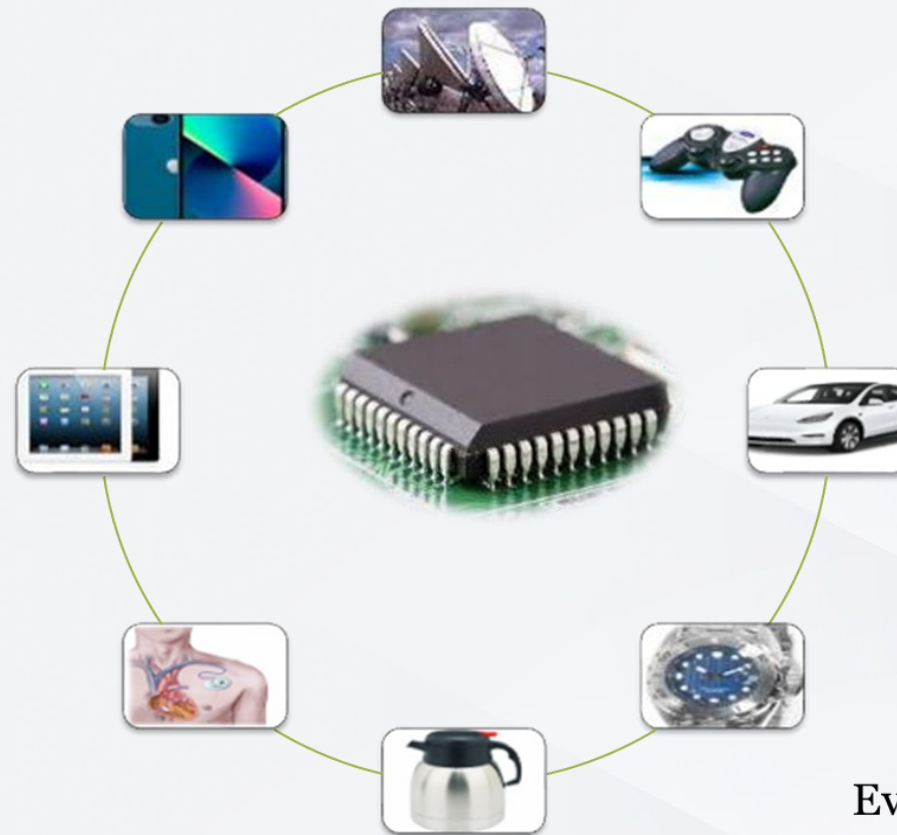




# Implementation of Semiconductor Chips

- Semiconductor chips are used everywhere:

- Computers
- Cellular phones
- Tablets
- Smart Phones
- Gaming systems
- DVD players, TVs
- Watches
- Cars
- Medical devices
- Pacemakers and coffee pots
- Space stations
- Greeting cards
- ...



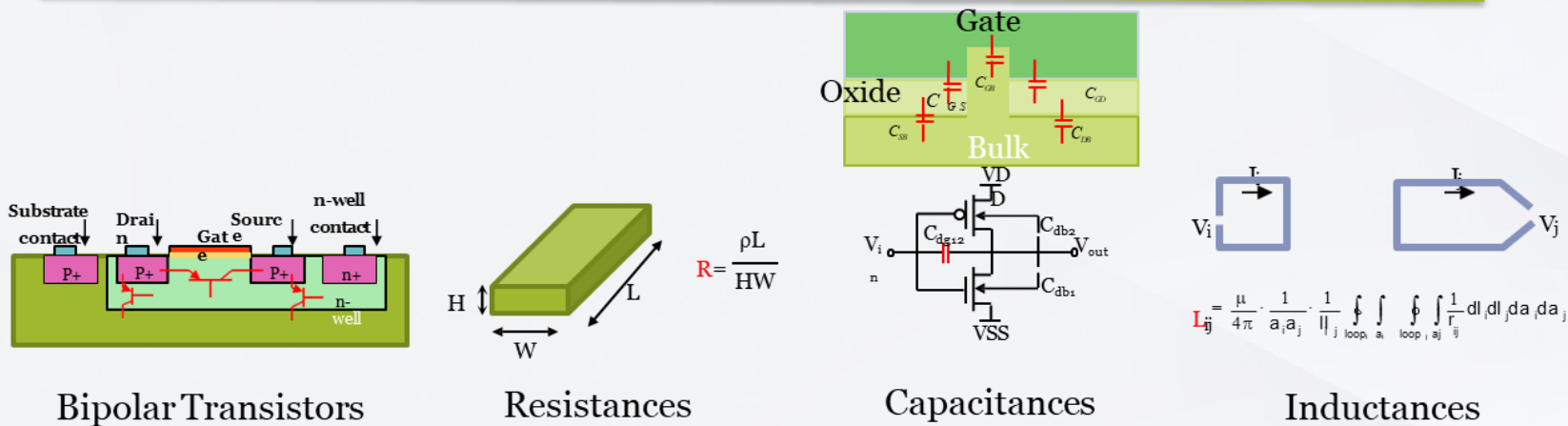
Everywhere

# Types of IC Elements

Useful

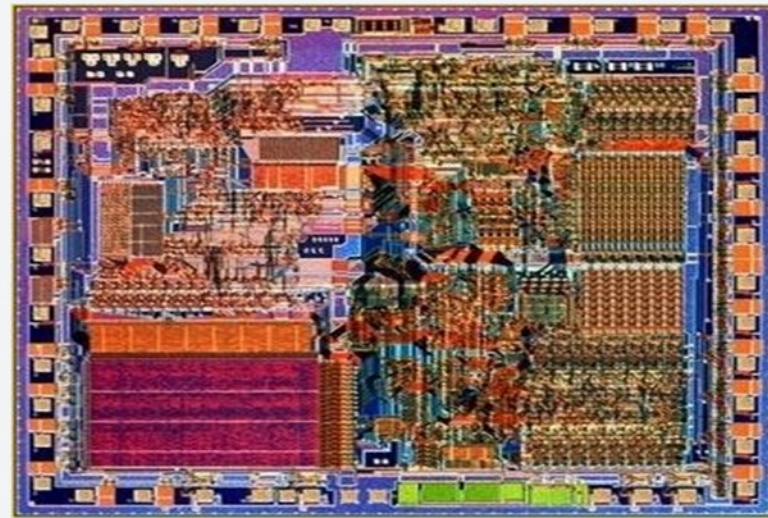


Parasitic



# IC Definition

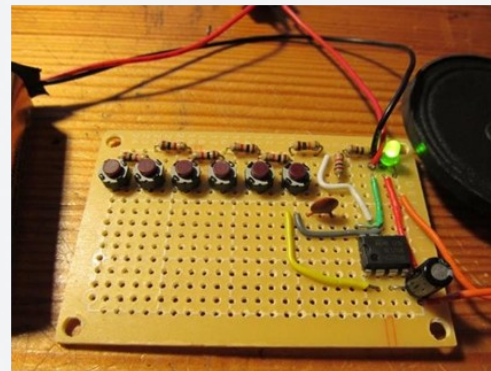
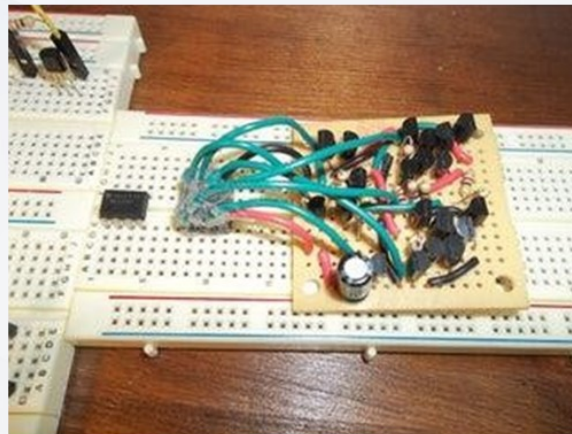
- Integrated circuits (IC) is a complex set of electronic components and their interconnections etched on a chip.



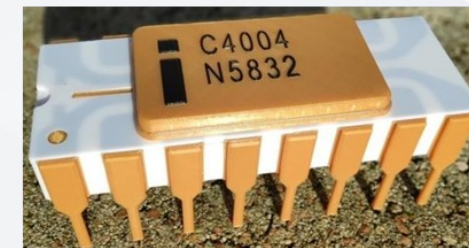
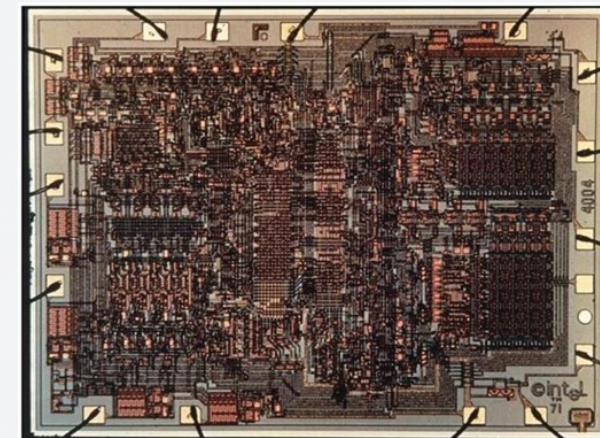


# Discrete vs. Integrated Electronics

Circuits using discrete components



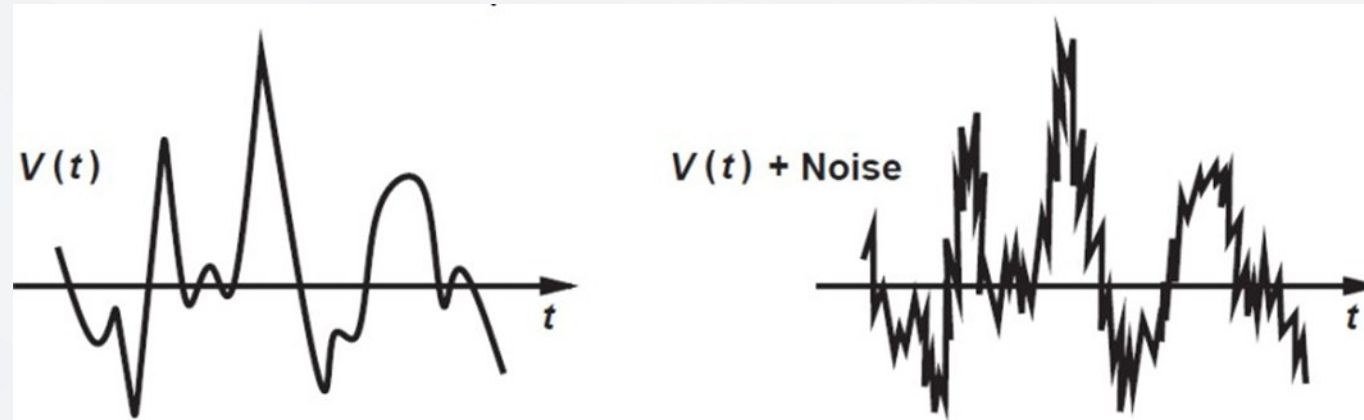
Integrated Circuits



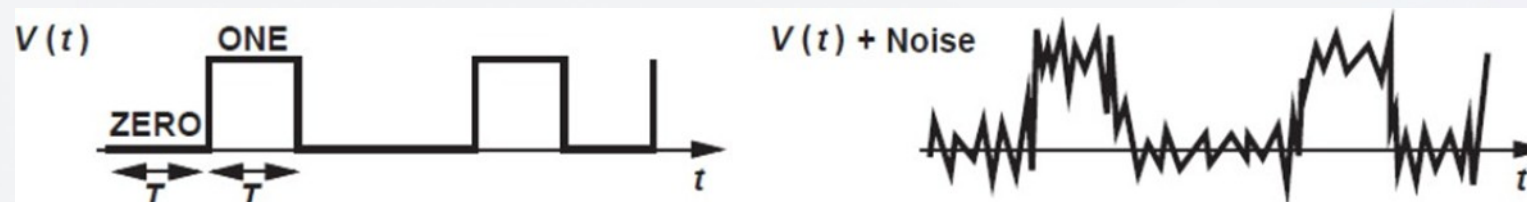


# Analog vs Digital Signals

□ Analog: continuous in time and amplitude



□ Digital: discrete in time and amplitude



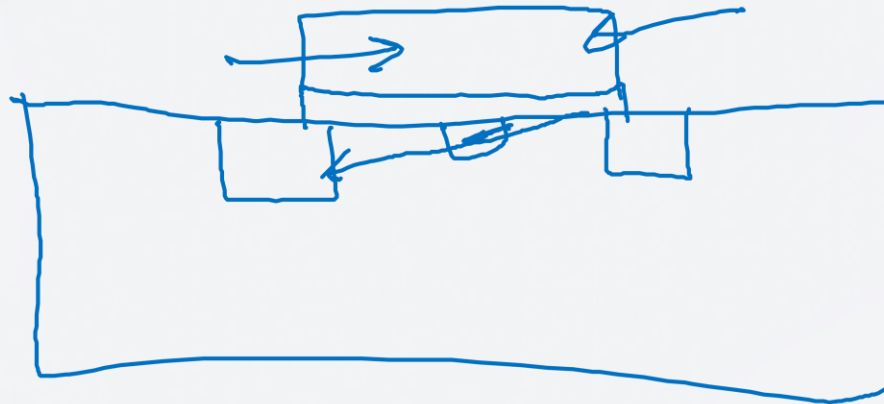
# Why Digital IC Design?

## □ Digital Circuits are:

- Less sensitive to noise (robust)
- Easier to store (digital memories)
- Easier to process (digital signal processing: DSP)
- Amenable to automated design
- Amenable to automated testing
- Easier to port from one technology to another
- Direct beneficiary of Moore's law (down-scaling)



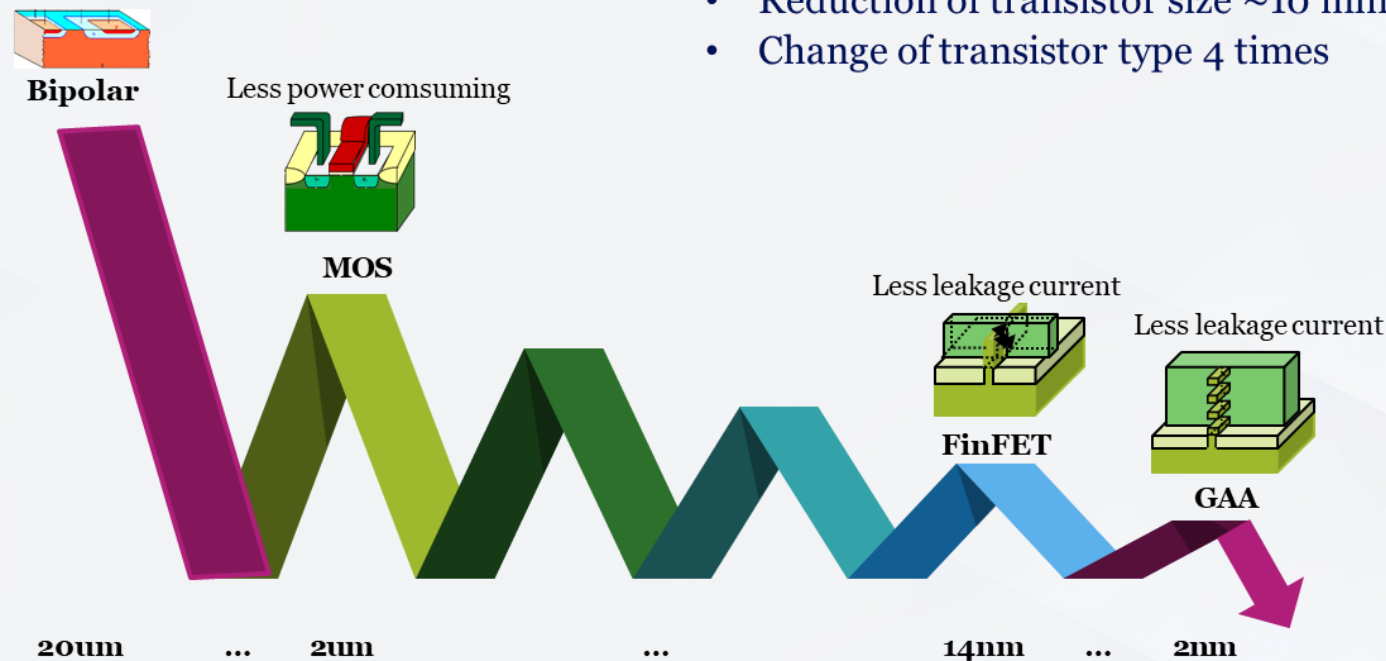
# IC Scaling



$$V_{th} = 0.7 \text{ V}$$

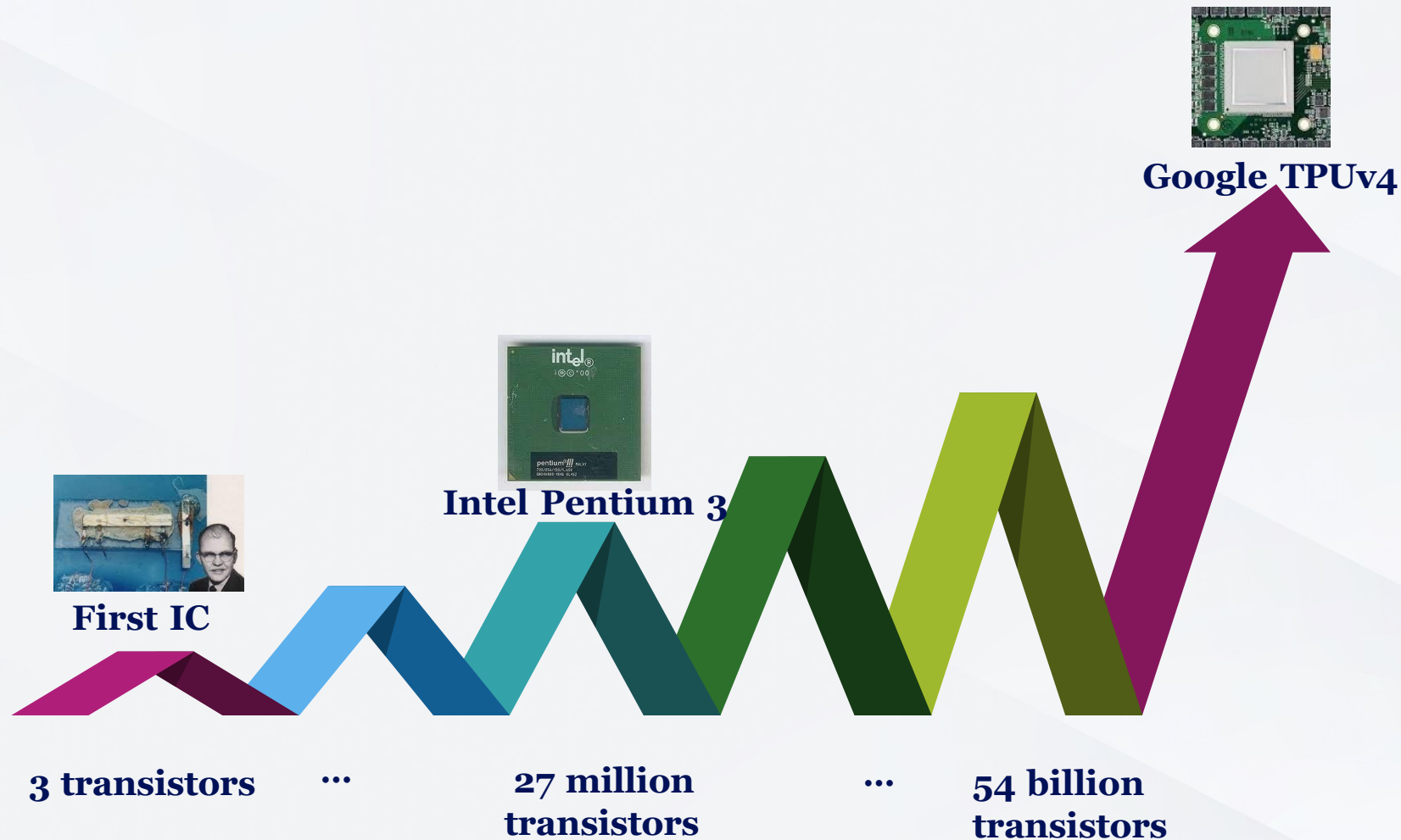
$$\sqrt{G_S} = 0.7$$

- Reduction of transistor size ~10 mln times
- Change of transistor type 4 times

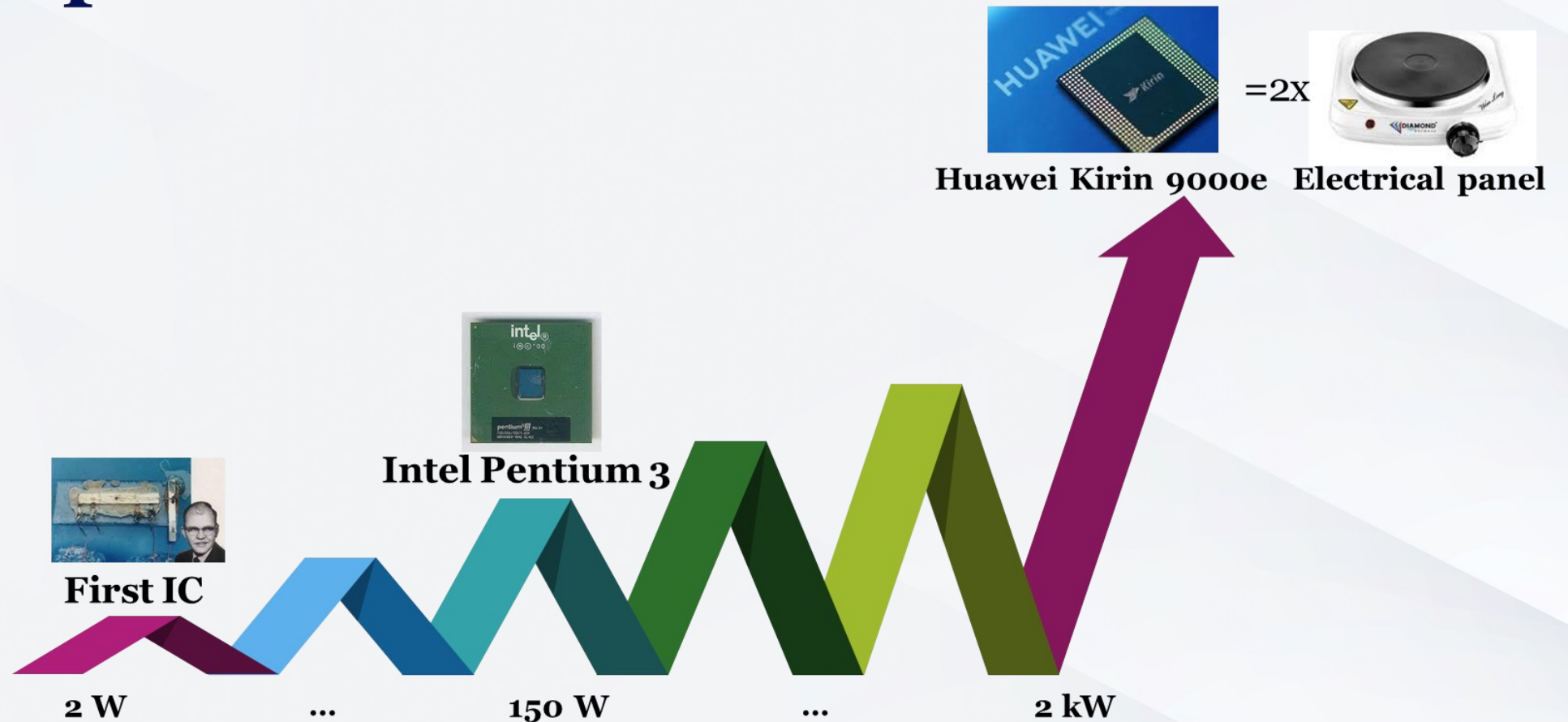




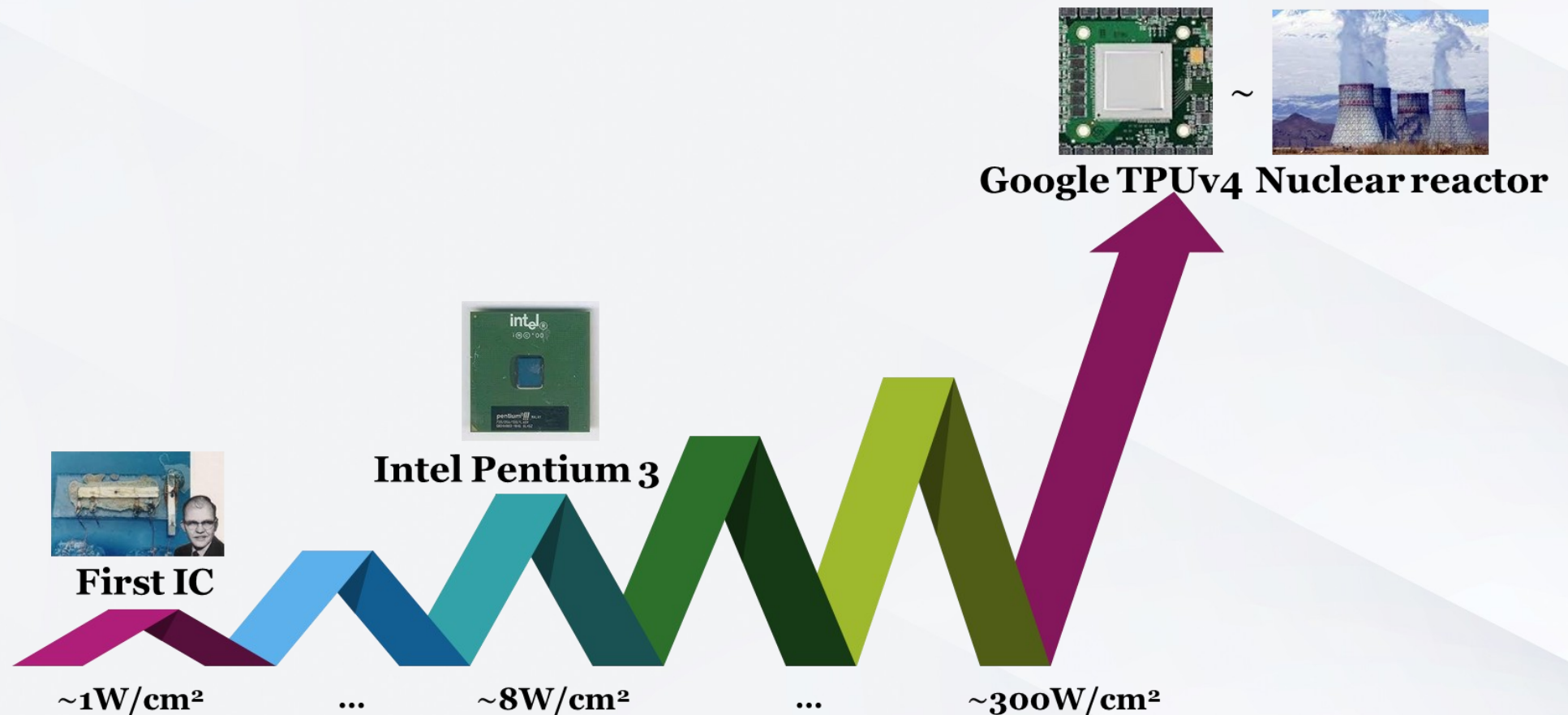
# Increase of IC Component Number



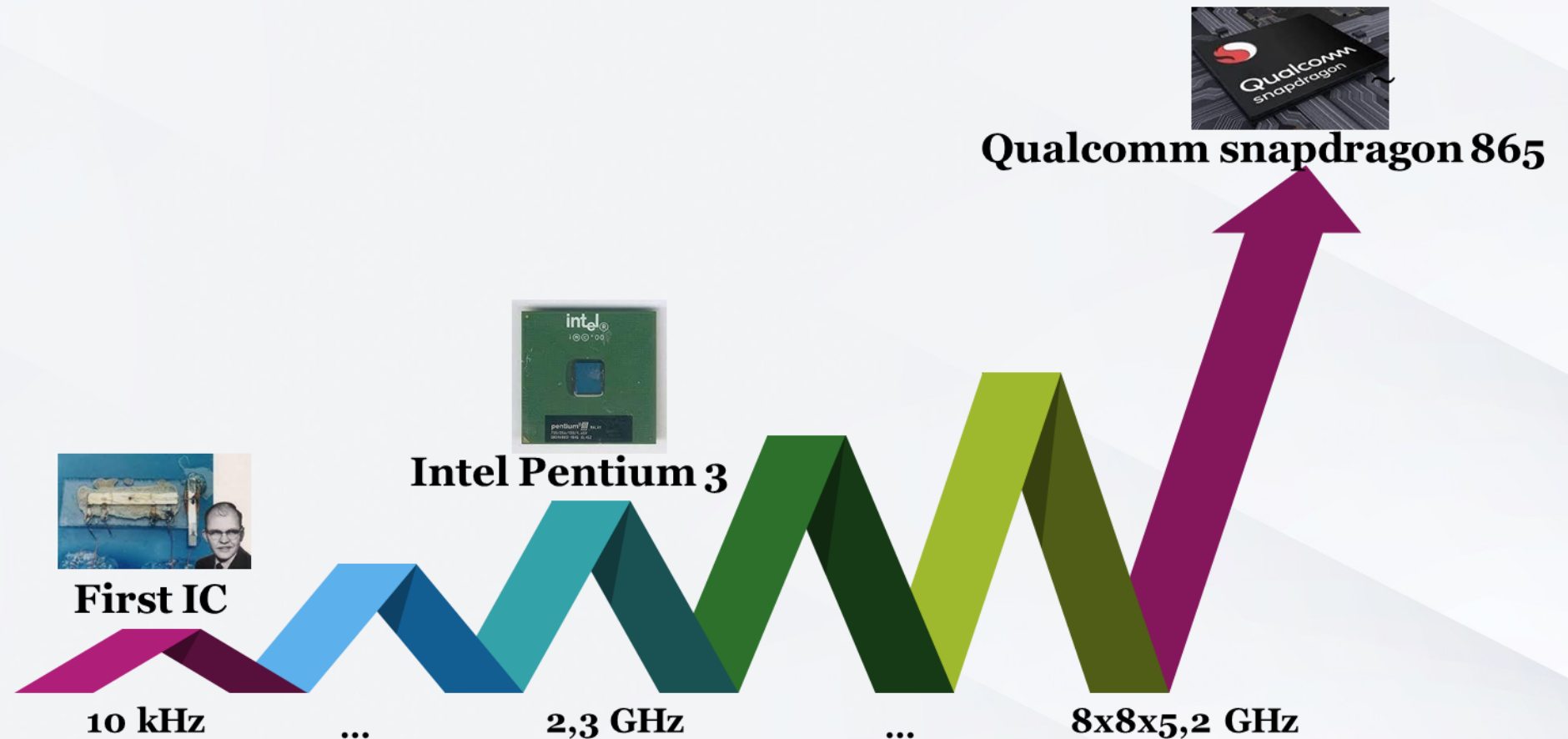
# Increase of IC Absolute Power Consumption



# Increase of IC Specific Power Consumption



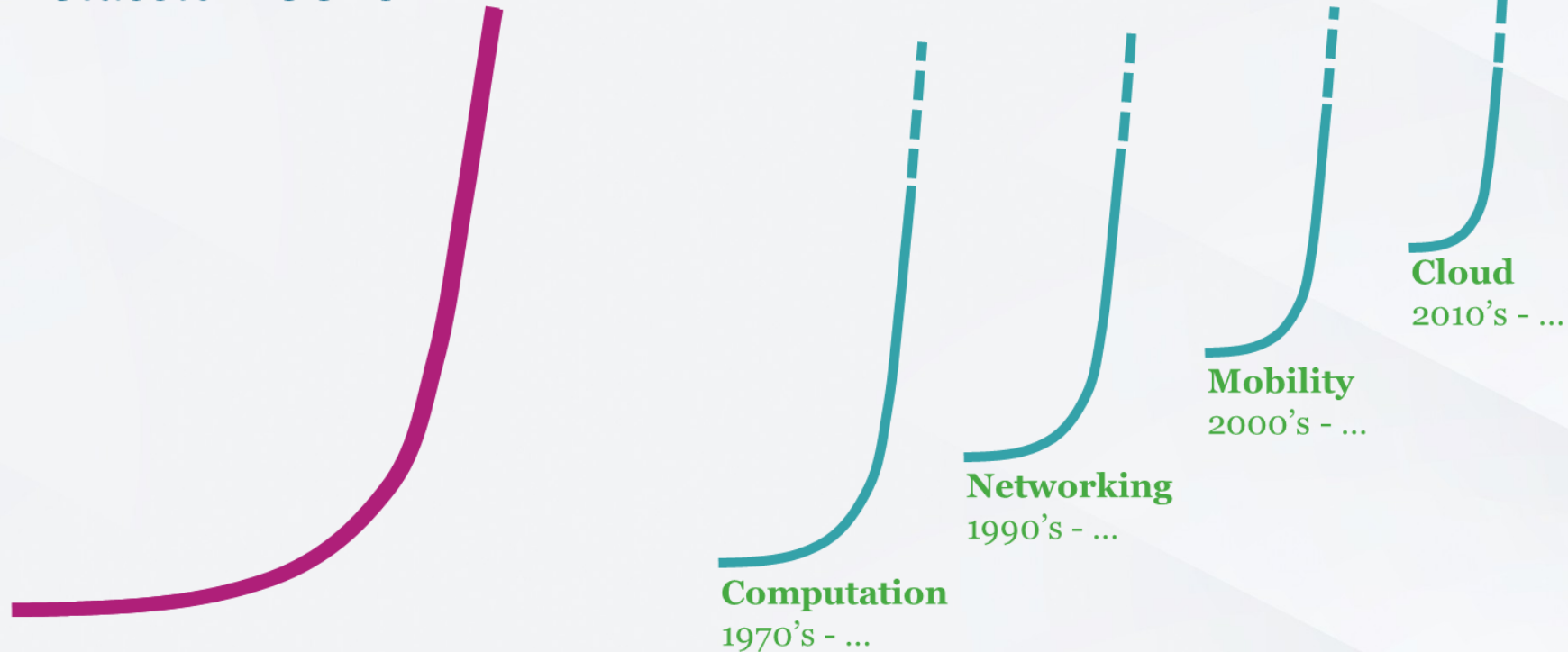
# Increase of IC Performance



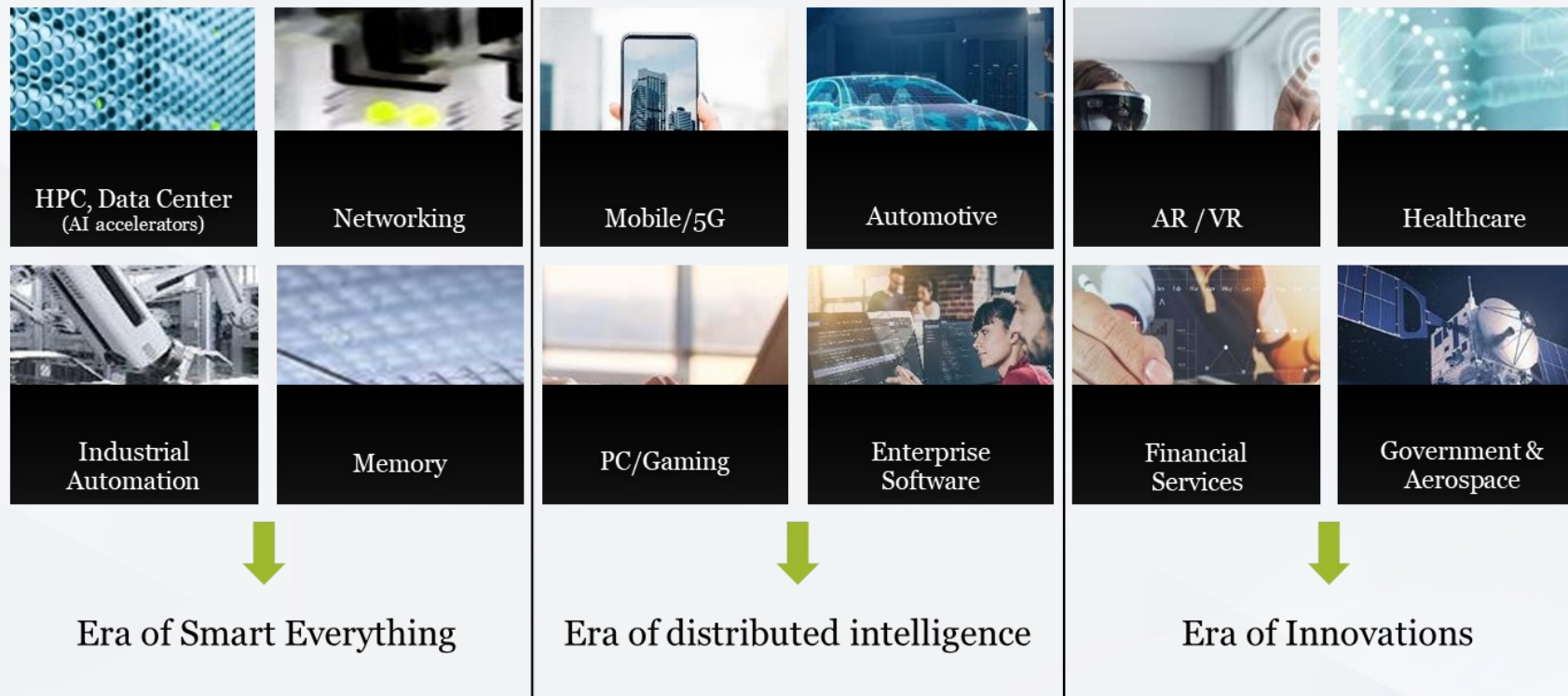


# Increase of Complexity

*Classic Moore*



# Changes in Markets



Most important: **automotive** and **IoT**

# Era of Smart Everything

From the everyday-helpful to the sublime

Voice Assistants



Setting up calendar...



Critical support  
for the elderly

Advanced Robotics



Home cleaning  
robots...



Remote robotic  
surgery

Mapping/Navigation



Package  
delivery...



Humanitarian  
mapping

Digital Imagery

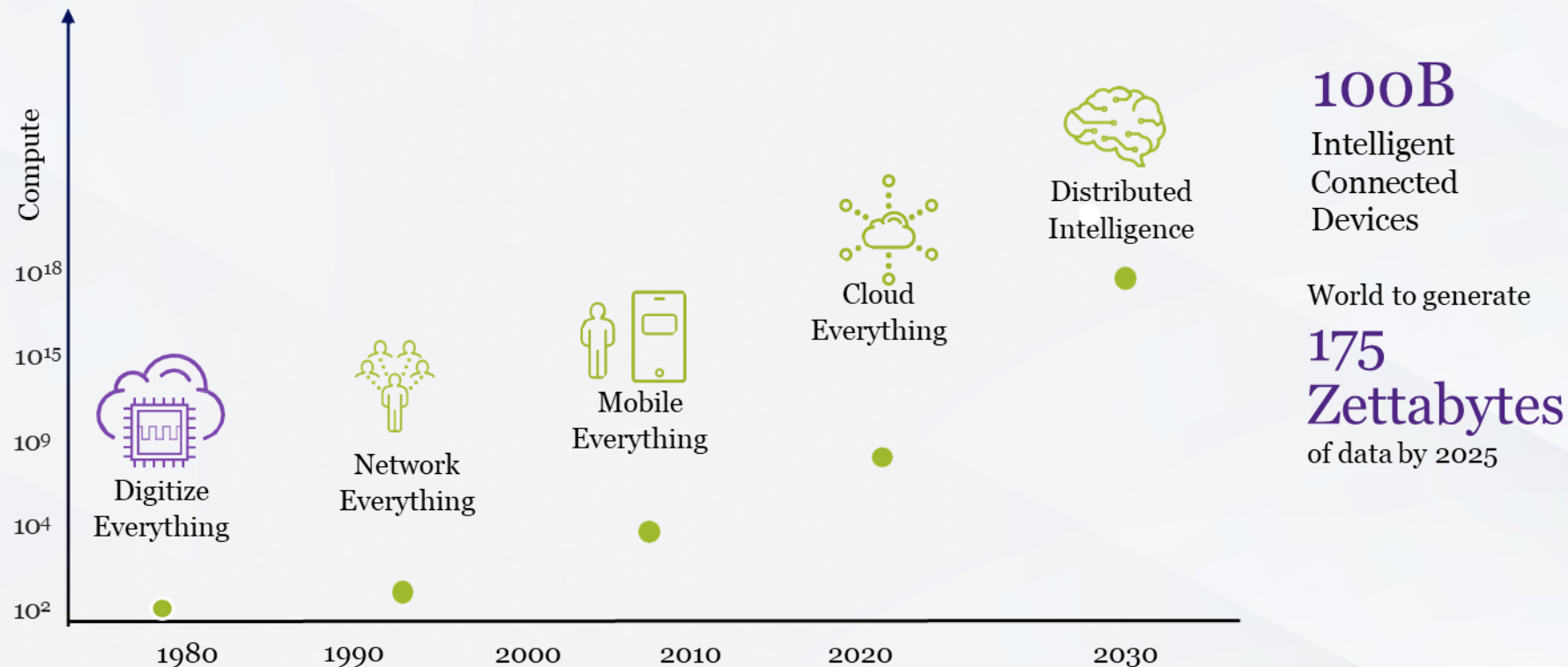


Smartphone photos...



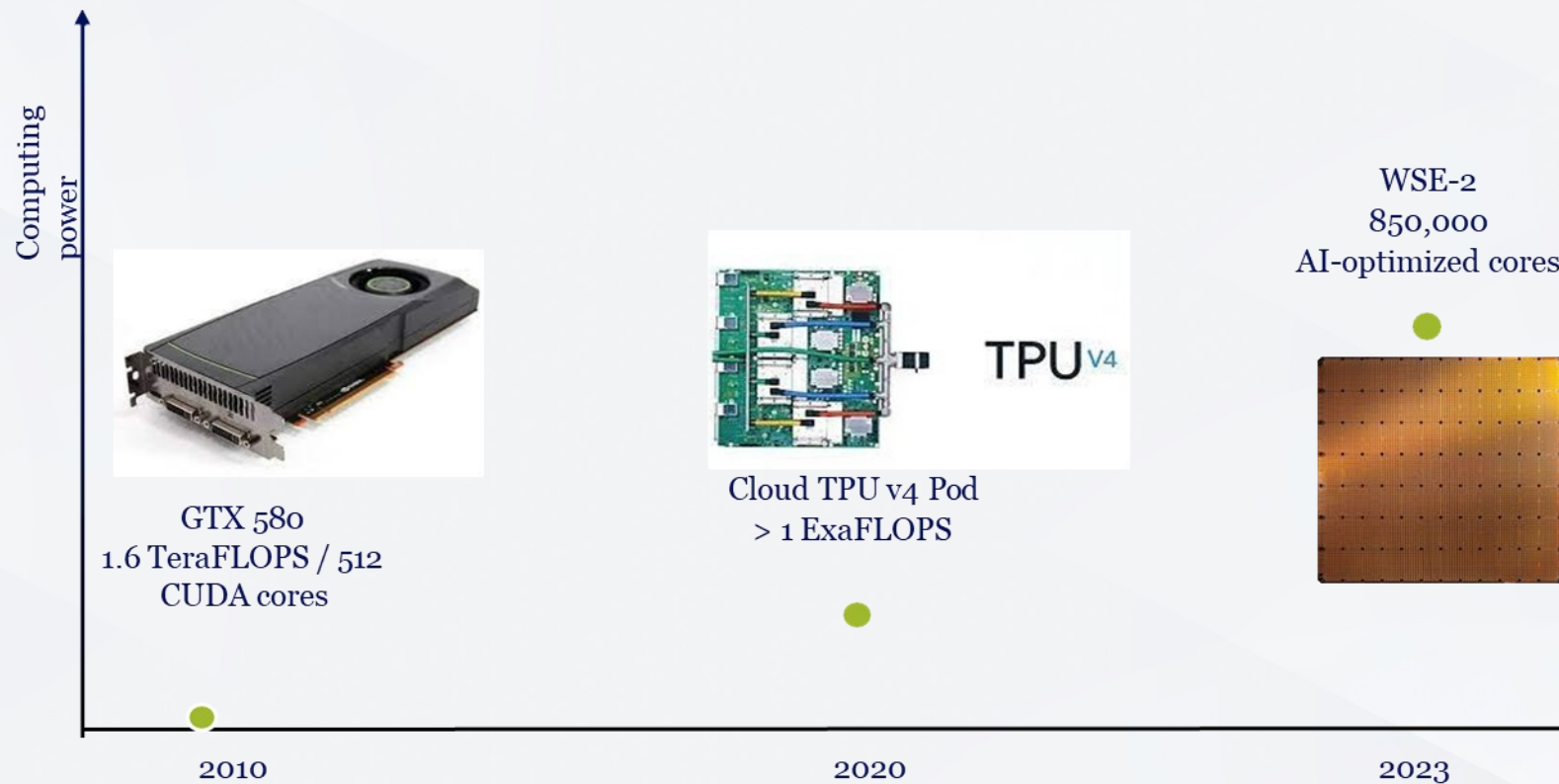
Diabetic retinopathy  
screening

# Era of Distributed Intelligence



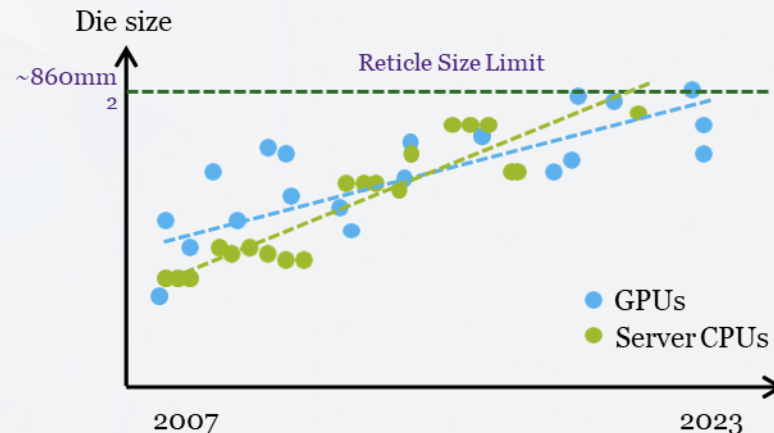


# Era of Innovations



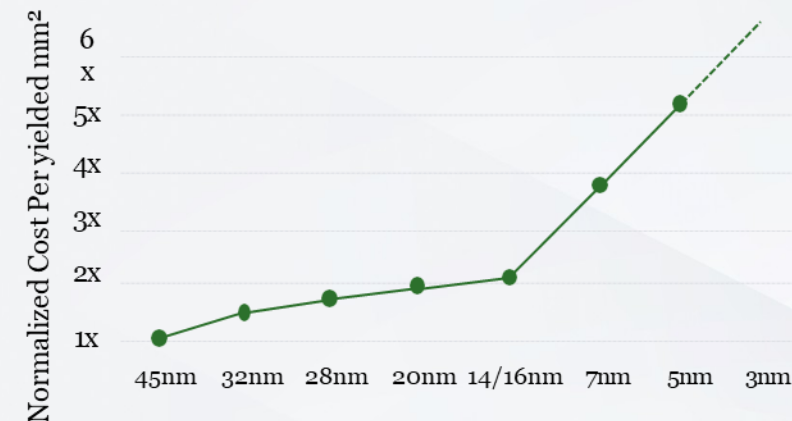
# Inconsistency between Moore's Law and the New Era

Compute performance reaching reticle size limit



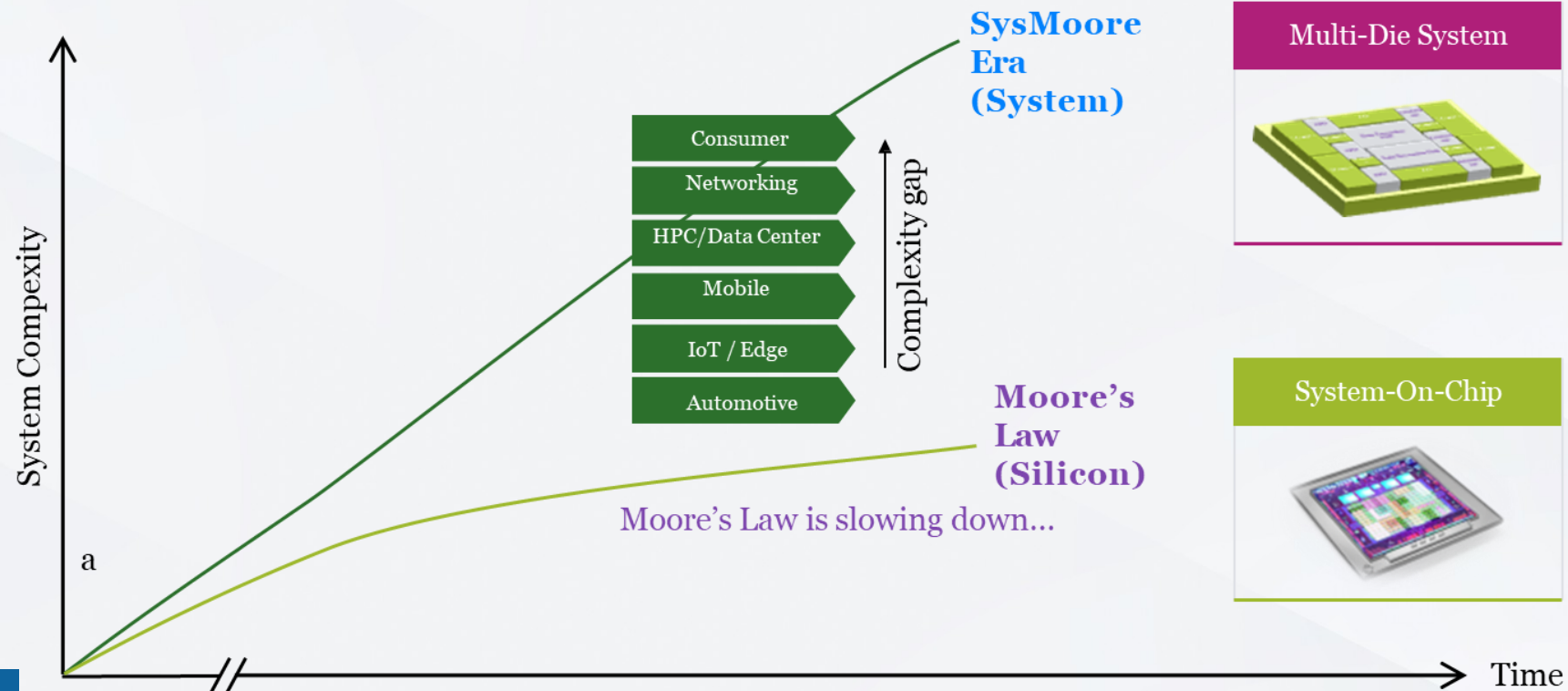
CPU/GPU complexity grew over the last 15 years

Cost of fabricating silicon accelerating

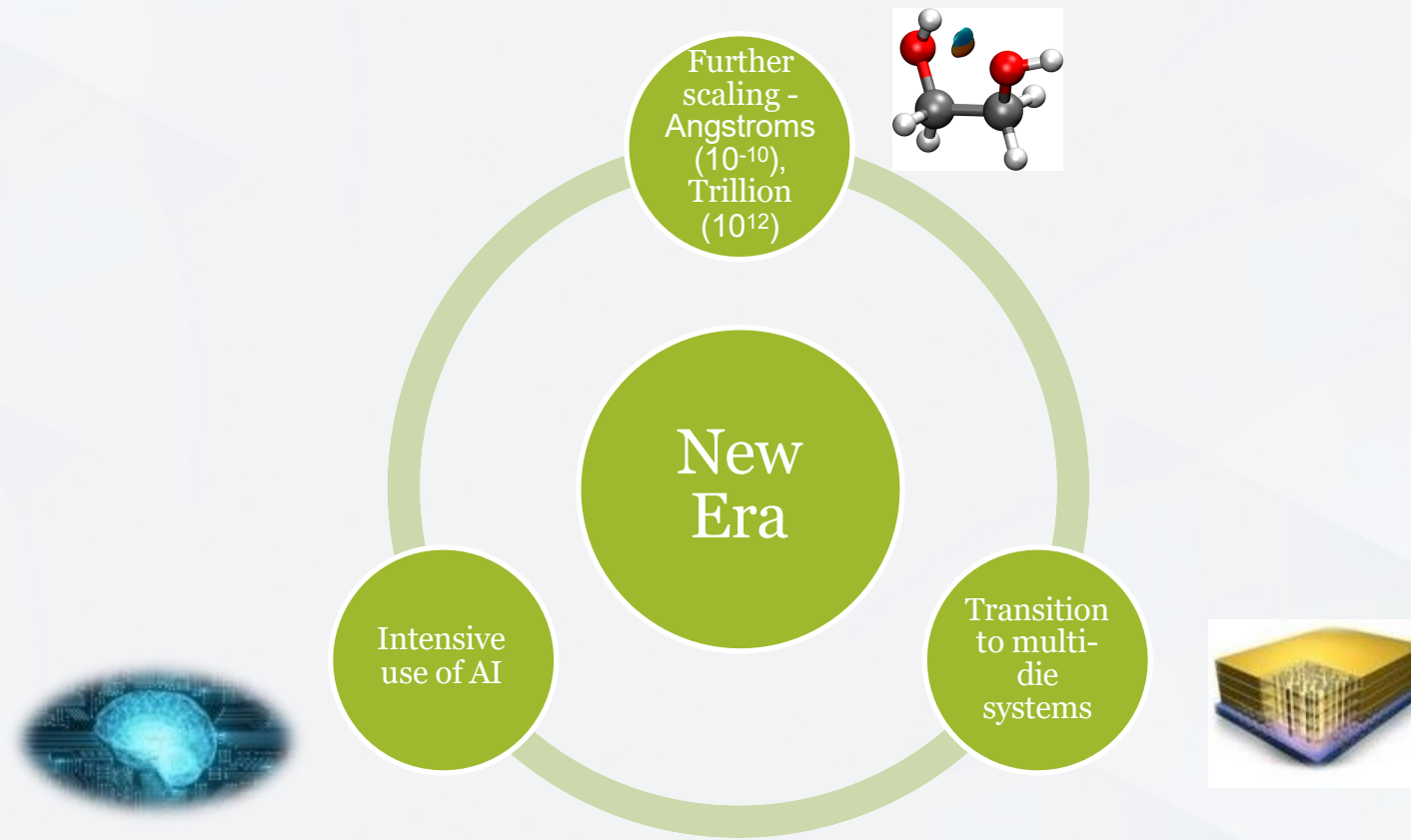


Silicon cost per yielded mm<sup>2</sup> for a 250mm<sup>2</sup> die

# Alignment with the New Era

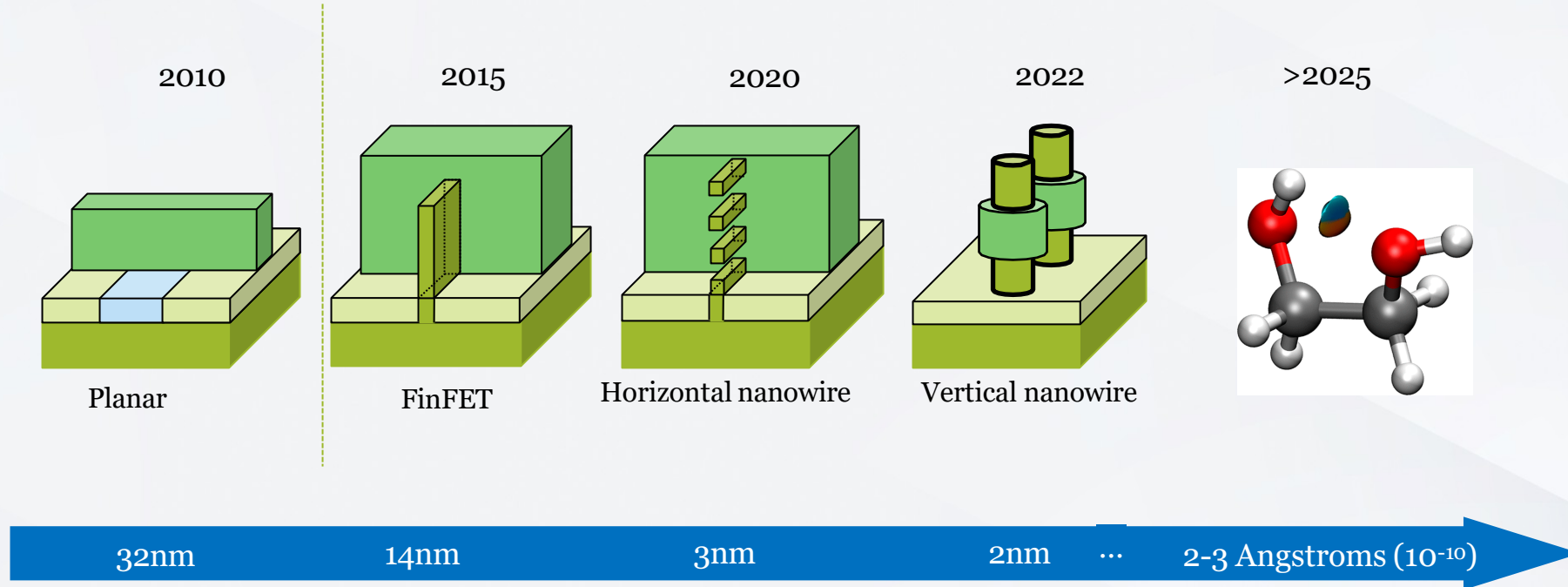


# Alignment with the New Era (2)

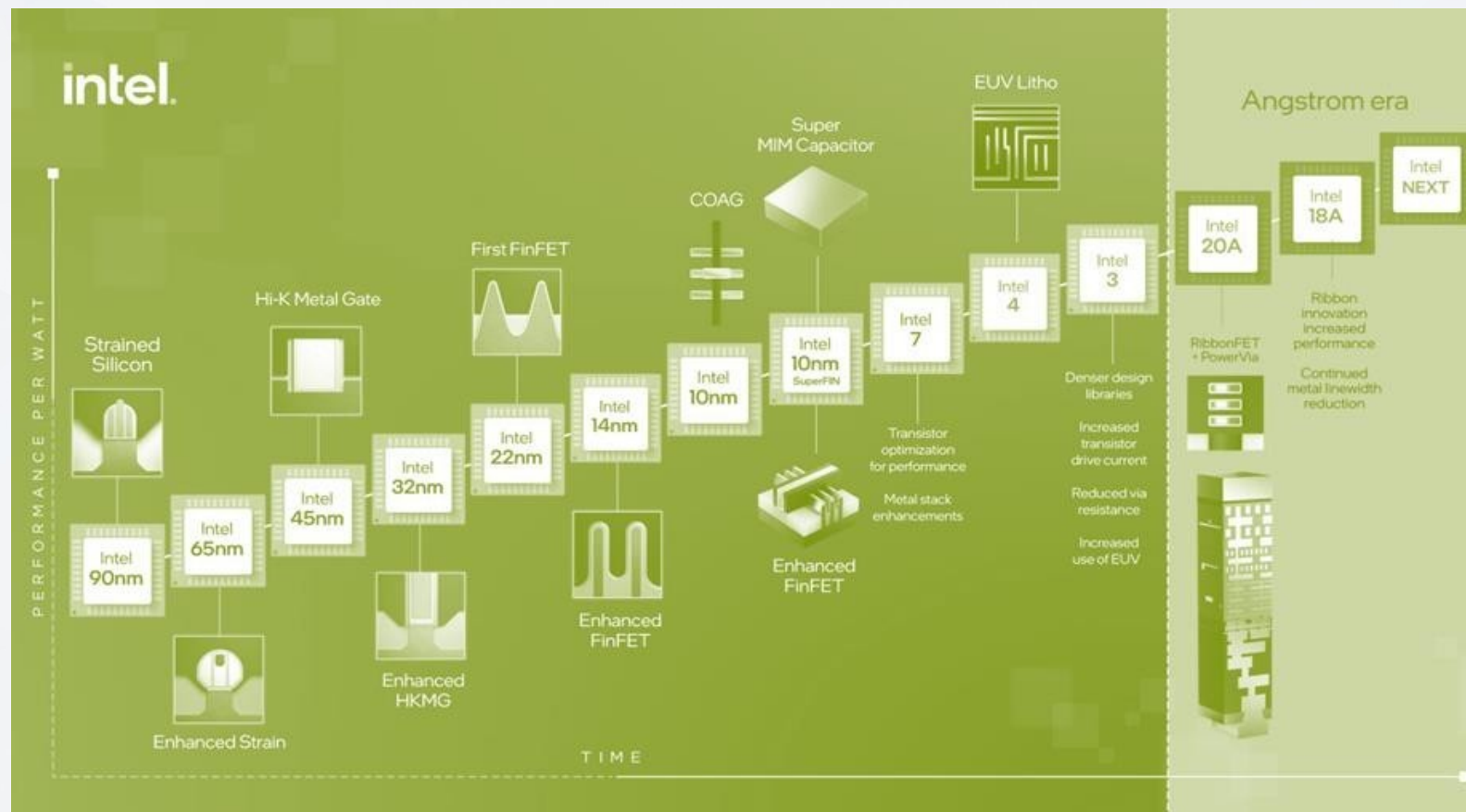




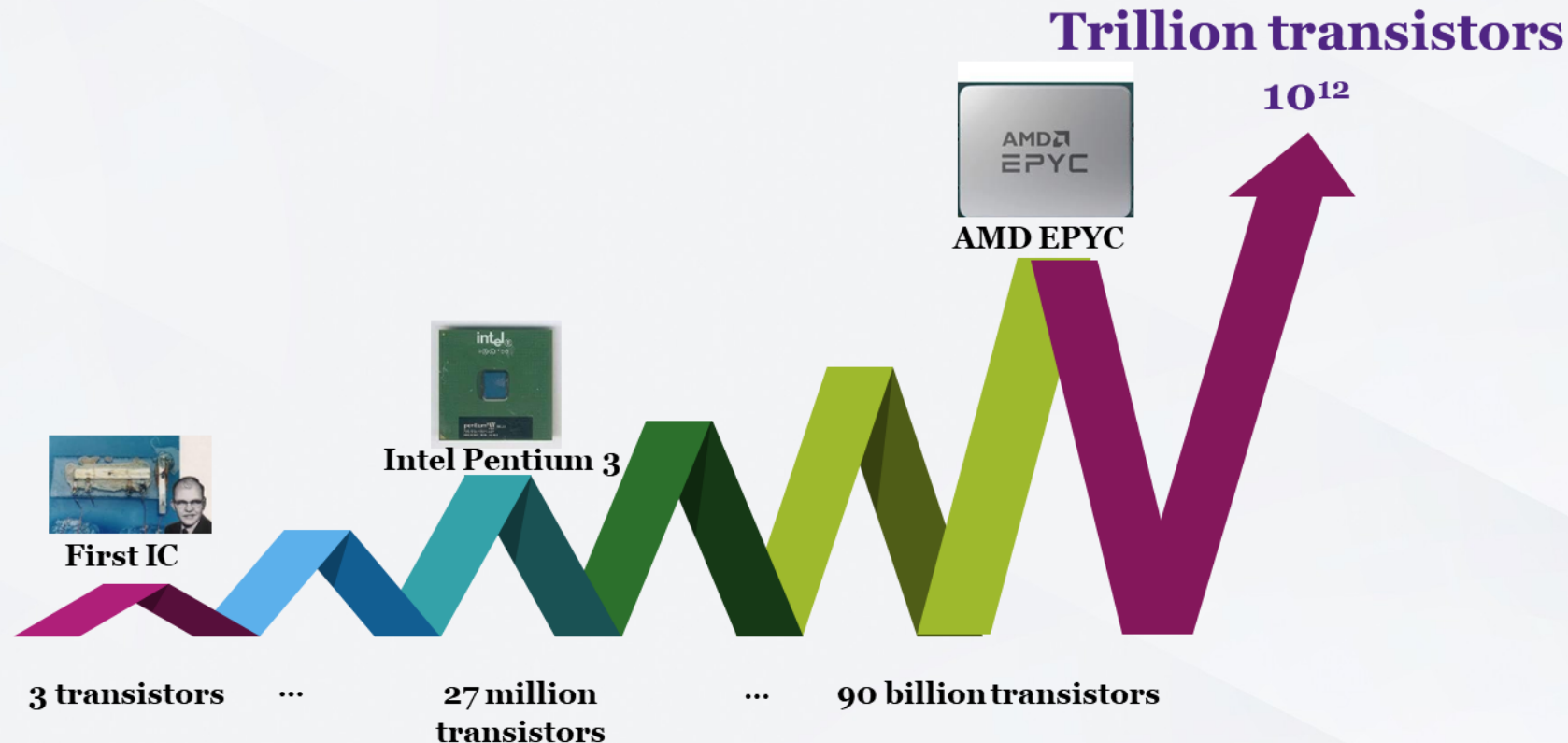
# Alignment with the New Era: Further Scaling



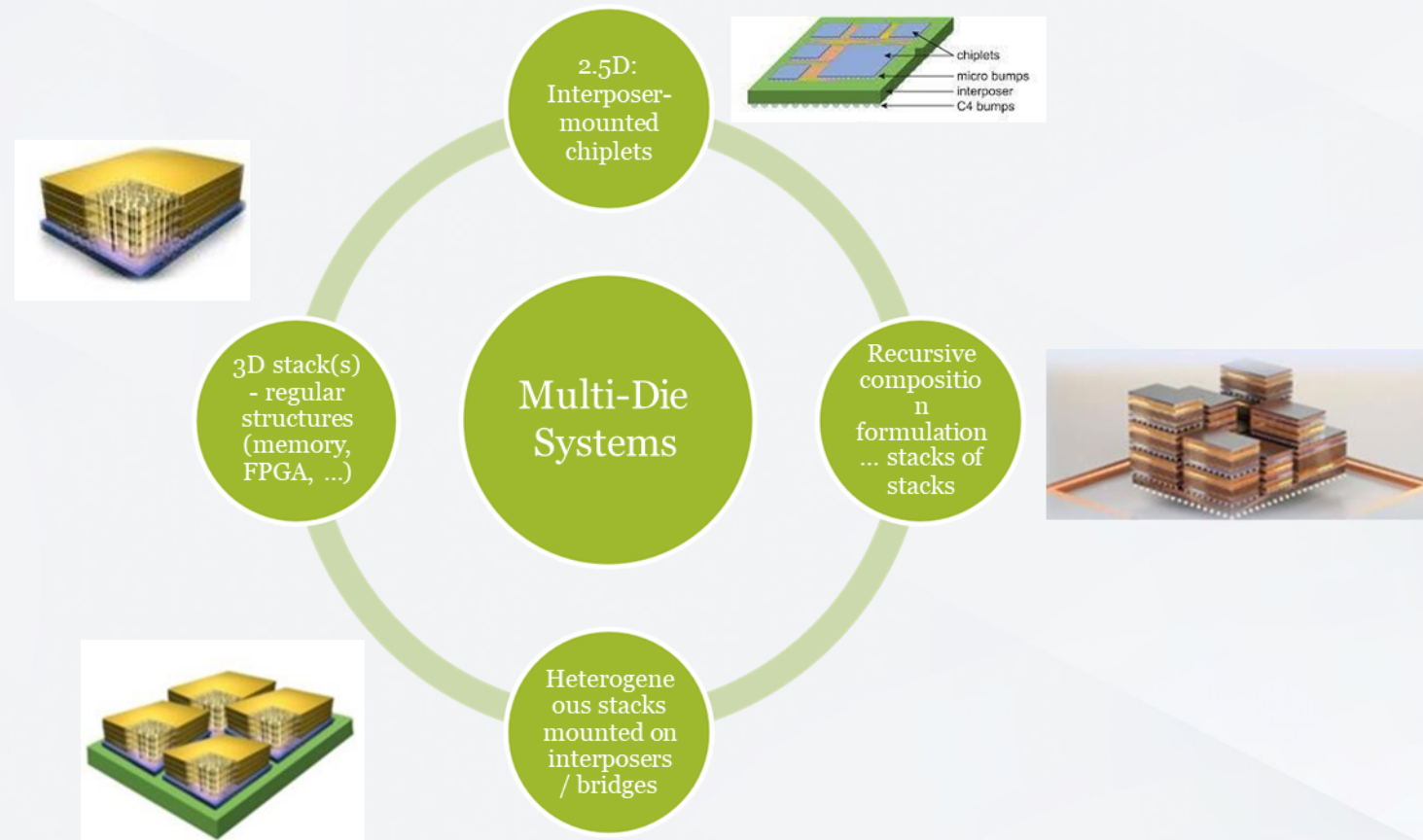
# Alignment with the New Era: Further Scaling (2)



# Alignment with the New Era: Further Scaling (3)

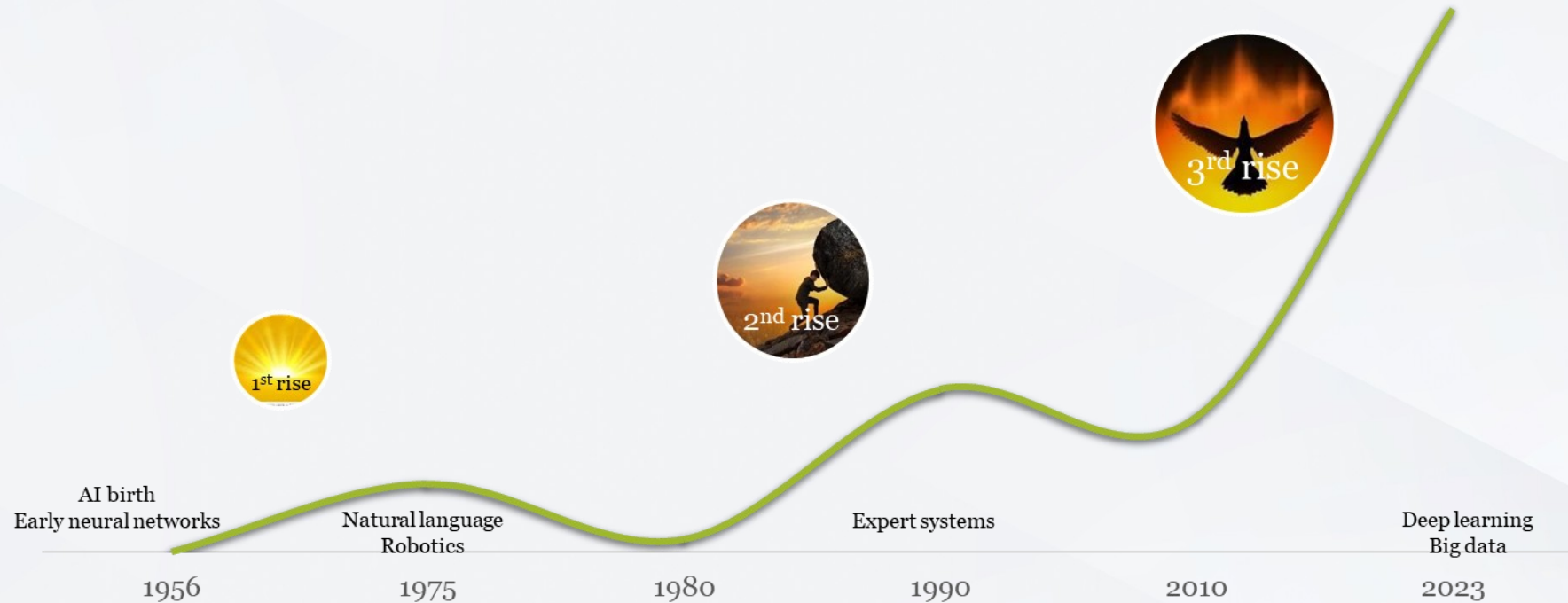


# Alignment with the New Era: Transition from Monolithic to Multi-Die Systems





# Alignment with the New Era: Intensive Use of AI



# History and Evolution of The IC Industry

- Semiconductor Industry Association (SIA) Roadmap

| Date                           | 1999    | 2005    | 2010  | 2018      | 2023      |
|--------------------------------|---------|---------|-------|-----------|-----------|
| Technology (nm)                | 180     | 65      | 28    | 5         | 1         |
| Minimum mask count             | 24      | 25      | 27    | 30        | 50        |
| Wafer diameter (mm)            | 200     | 400     | 400   | 450       | 500       |
| Memory samples (bits)          | 1G      | 8G      | 32G   | 10T       | 100T      |
| Transistors/cm <sup>2</sup>    | 6.2M    | 180M    | 330M  | 1.5G      | 20 G      |
| Maximum number of metal layers | 6-7     | 9       | 9     | 12        | 25        |
| Clock frequency (MHz)          | 1250    | 3200    | 5200  | 20000     | 35000     |
| IC sizes (mm <sup>2</sup> )    | 400     | 596     | 699   | 750       | 1200      |
| Power supply (V)               | 1.5-1.6 | 0.8-1.2 | 1.2-1 | 0.37-0.42 | 0.28-0.33 |
| Maximum power (W)              | 90      | 150     | 171   | 183       | 2200      |
| Number of pins                 | 700     | 1957    | 2734  | 3350      | 4200      |

# Technology Roadmap



# VLSI Digital Design Style

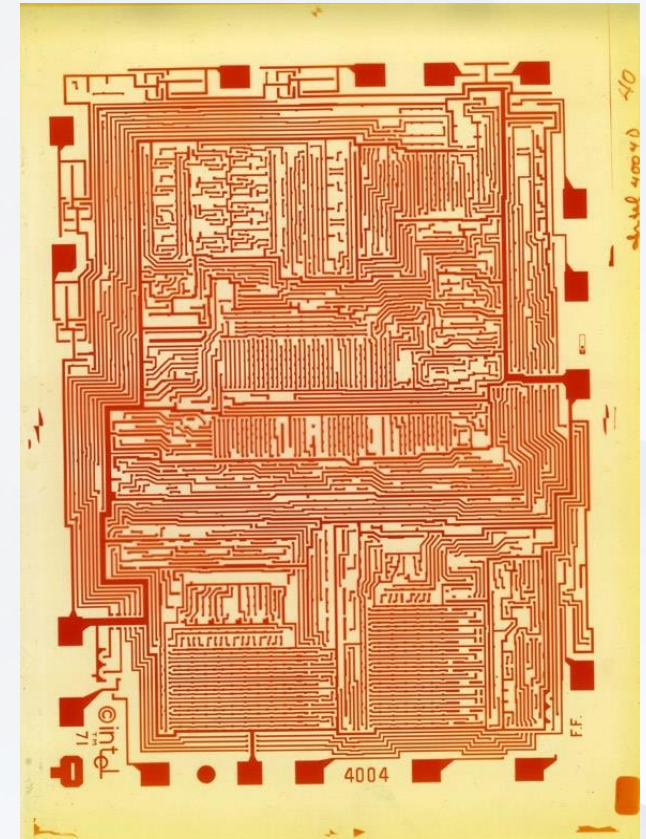
- ❑ **Due to time and cost constraints, very few teams and/or companies develop products from device level through to system level.**
- ❑ **Various Design Styles are available to shorten the time-to-market and development cost.**
- ❑ **Trade-offs are taken into consideration, as abstractions are usually designed generically and therefore come with some overhead.**
- ❑ **In the following slides, we will briefly discuss:**
  - Full custom design
  - Standard Cell based ASIC design
  - Field-Programmable Gate Array (FPGA) design
  - Microprocessor (Software)



# VLSI Design Styles – Full Custom

## □ Full Custom Design

- The original design style.
- Everything is done at transistor level
- Rarely used in digital design
- High cost
- But gives significant gain in performance.
- Analog designs mostly



# VLSI Design Styles – Semi Custom

- **Design is compiled using ready blocks:**

- Technology-dependent fixed size physical designs

Digital standard cells

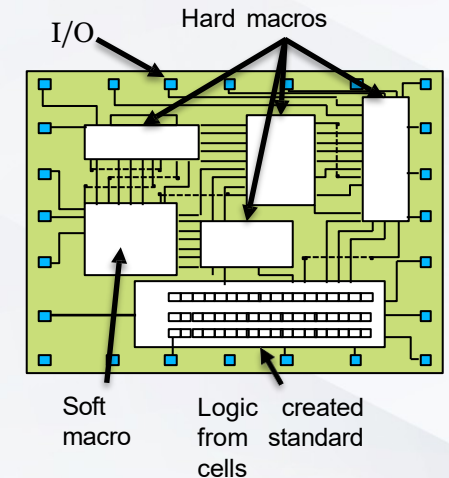
Small cells performing simple logic used to build larger designs

(Boolean primitives: and, or, xor; flip-flops; adders, etc.)

Hard macros (Hard IP)

Ready physical design (analog cells, I/O cells, etc.)

- Technology-independent behavioural descriptions, synthesizable to physical design using standard cells and if necessary hard-IPs



# VLSI Design Styles – Semi Custom

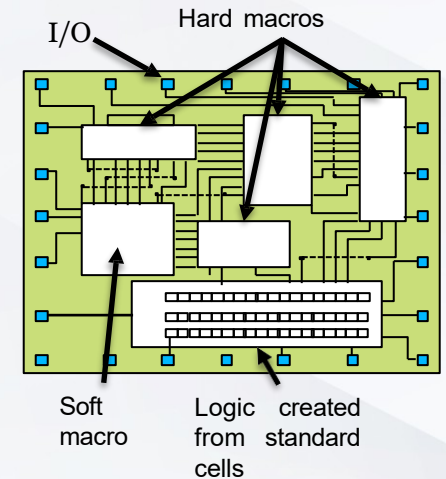
- **Design is compiled using ready blocks:**
  - Technology-dependent fixed size physical designs

Soft macros (Soft IP)

**Synthesizeable register transfer level descriptions of complex functions (processor cores, arithmetic units, etc.).**

System-level macros (SLMs)

**High level behavioural descriptions (DSP logic, digital filters, etc.)**



# VLSI Design Styles - FPGA

## □FPGA – Field Programmable Gate Array

- Array of configurable logic blocks, and programmable interconnect structures
- Fast prototyping, cost effective for low volume production
- HDL (hardware description language) – is used





# Design/Manufacturing Options

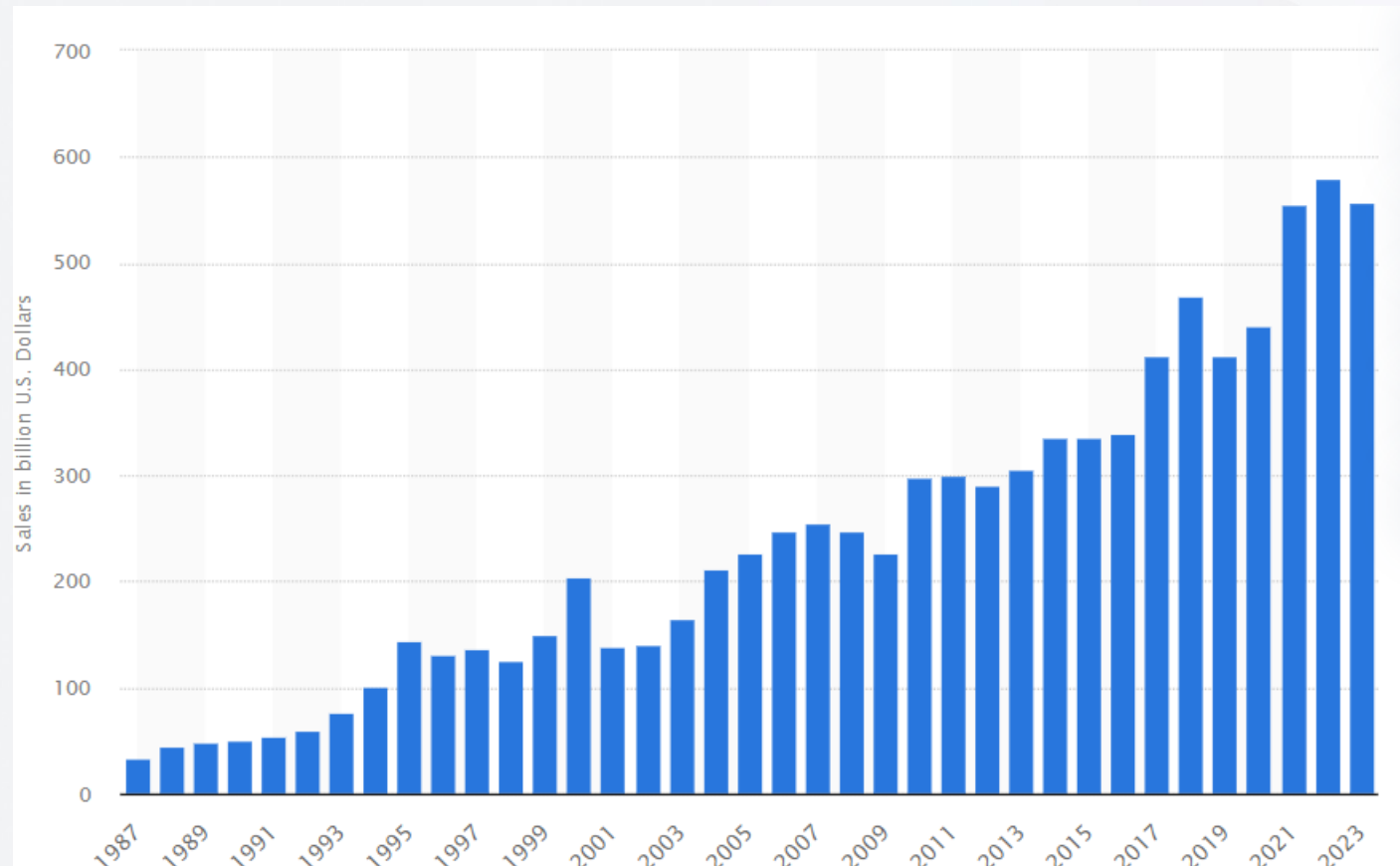
- **Full-Custom**
  - All design and manufacturing process cycles are circuit specific
- **Semicustom**
  - Some design and manufacturing cycles is predefined
- **Programmable**
  - Functionality achieved by configuring (programming) already fabricated general purpose IC

|                                     | Full-Custom | Semi-Custom | Prog. |
|-------------------------------------|-------------|-------------|-------|
| Cost<br>(design/manufacturing)      | High        | Small       | Low   |
| Quality<br>(performance/area/power) | Best        | Low         | Low   |
| Time-to-Market<br>(time-to-market)  | Long        | Short       | Short |
| Production Volume<br>(use cases)    | Large       | Medium      | Small |

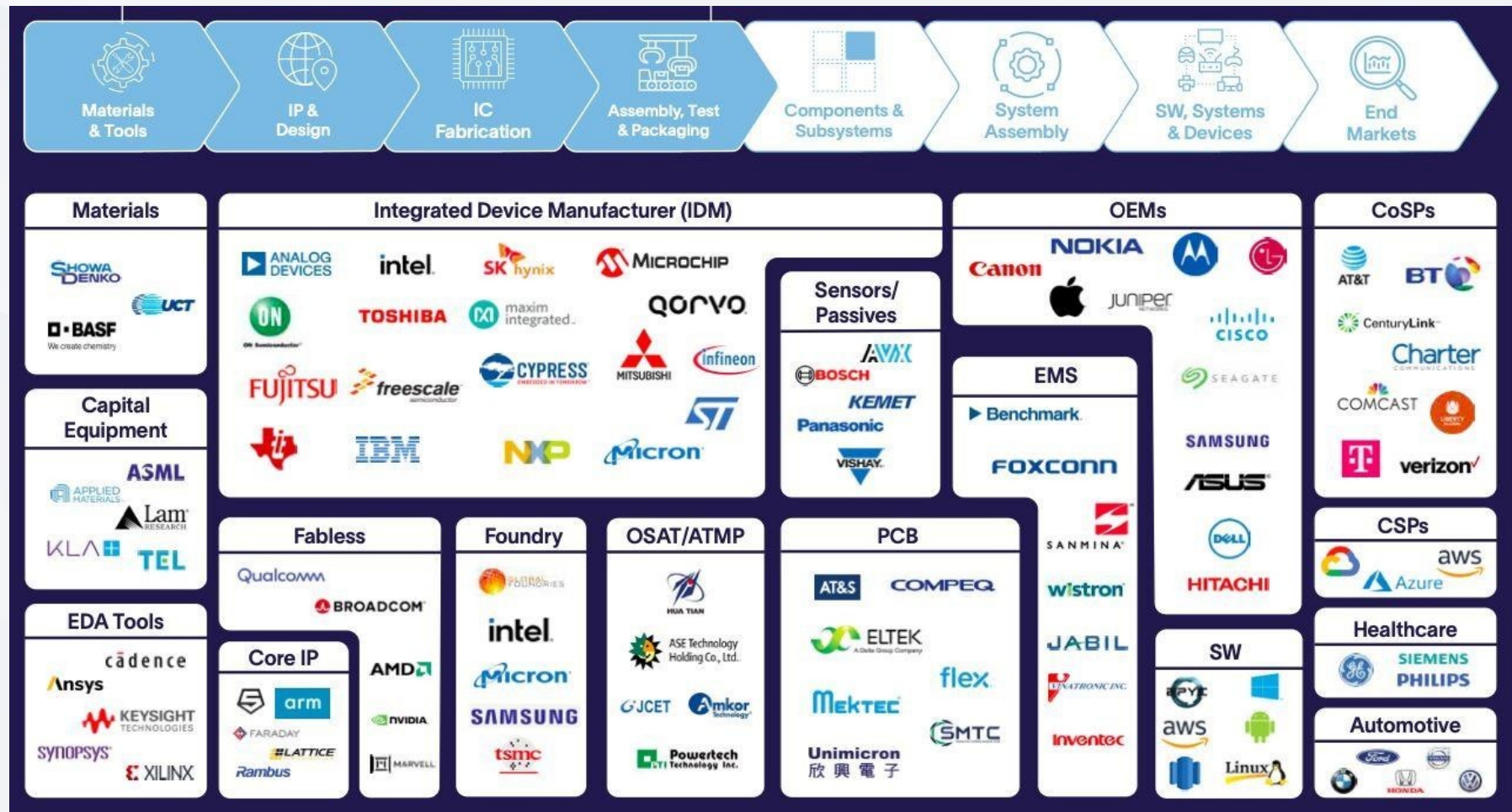
Full-Custom and Semicustom fabrication is used for Application Specific ICs (ASIC), i.e. manufactured for specific purpose

# IC Industry overview

- ❑ Semiconductor sales revenue worldwide (2022)
  - ~ 600 billion USD



# VLSI Design Ecosystem



# IC Industry in Egypt





# IC Fabrication process

❑ Sand to Silicon process:

GLOBALFOUNDRIES Sand to Silicon

# The Life of CMOS Inverter

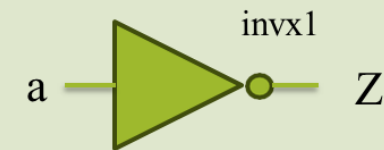
## Specification

“Device that outputs the inverse of its input with minimum size and power”

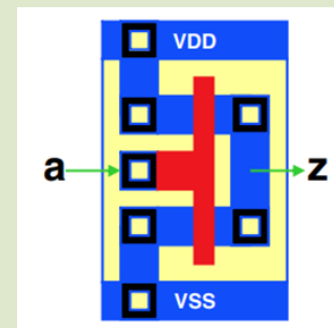
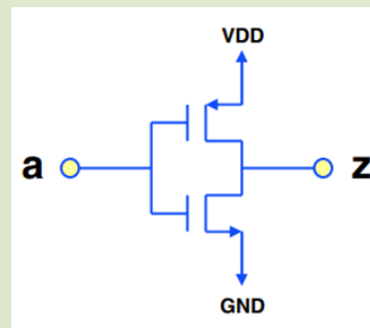
## RTL

```
module invx1(
  input wire a,
  output wire z );
  assign z=!a;
endmodule
```

## Gates

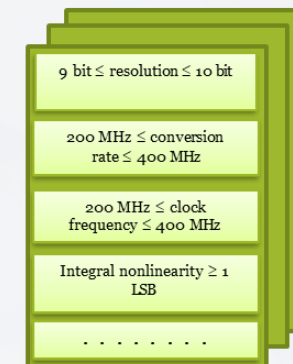


## Transistor and Layout

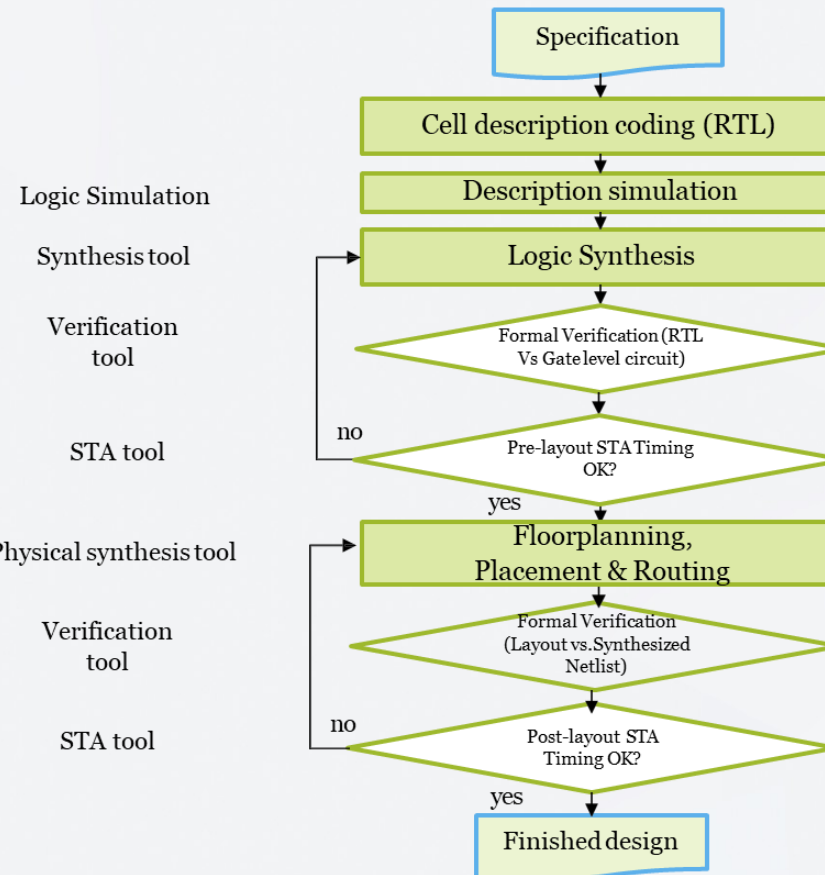
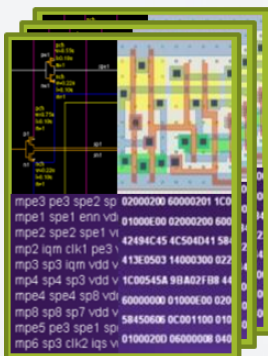


# Digital Design Flow

## Specification



## IC design



# Digital IC Designer

## □ How to be a Digital IC Designer?

- knowledge of Logic Design basics and concepts
- knowledge of Computer Organization and Architecture
- Very good understanding of Hardware Description language HDLs
- Very good understanding of Digital IC Design flow
- Hands-on experience using Digital design **Synopsys/Cadence** tools
- Scripting languages (Python- TCL- Perl.....)
- Basic knowledge of IC industry with all division, companies and customers.
- Logical thinking and creativity for problem solving



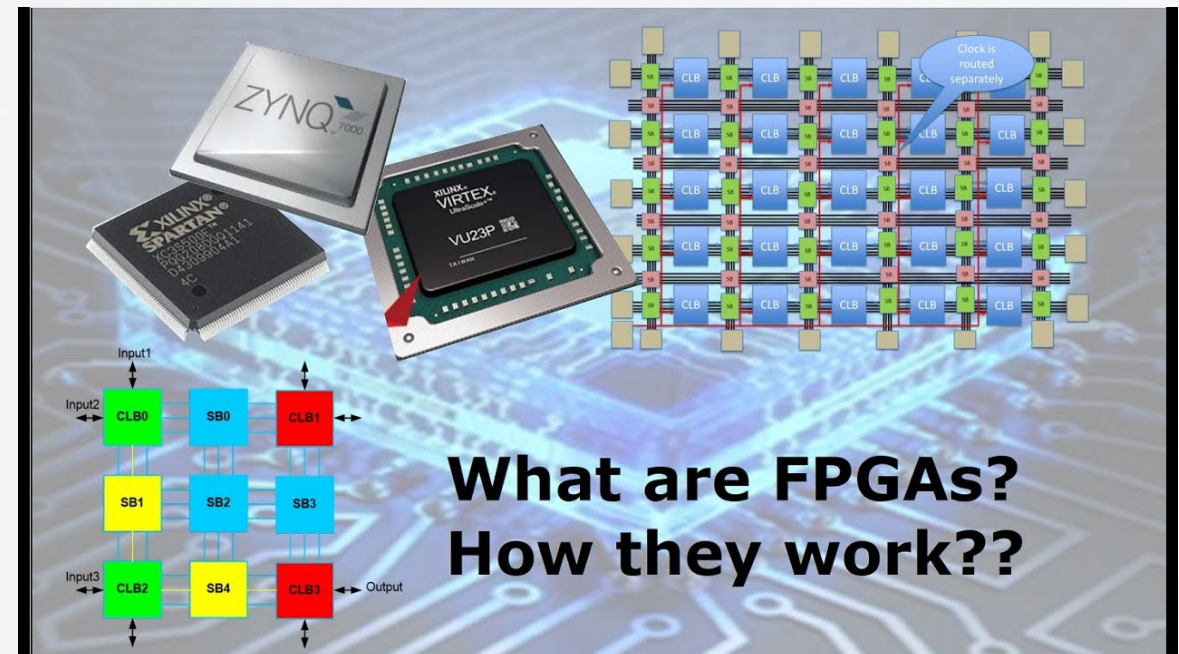
# Digital ASIC Design Flow Responsibilities: Frontend VS Backend

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Digital Frontend team</b></p> | <ul style="list-style-type: none"> <li>• Choosing microarchitecture, suitable algorithms to use, number of pipeline stages...etc according to the system specifications.</li> <li>• Developing synthesizable RTL and constraints.</li> <li>• Developing suitable DC scripts for synthesis and DFT.</li> <li>• Specifying power domains and generating UPF maps.</li> <li>• Performing verification for the RTL.</li> <li>• Performing gatelevel simulations to check backend deliverables.</li> </ul>                                                                                                                  |
| <p><b>Digital Backend team</b></p>  | <ul style="list-style-type: none"> <li>• Meeting timing requirements of setup and hold according to SC library specifications.</li> <li>• Meeting special timing requirements asked by the digital team (ex. Skew balancing).</li> <li>• Matching digital team intended design: gatelevel netlist matching RTL.</li> <li>• Meeting physical design rules specified by the foundry (DRC, LVS, Antenna, DFM)</li> <li>• Minimizing IR drop over the design so that it is below a defined threshold (~2%)</li> <li>• Minimizing power consumption so that it's comparable to a similar node or similar design.</li> </ul> |

# FPGA

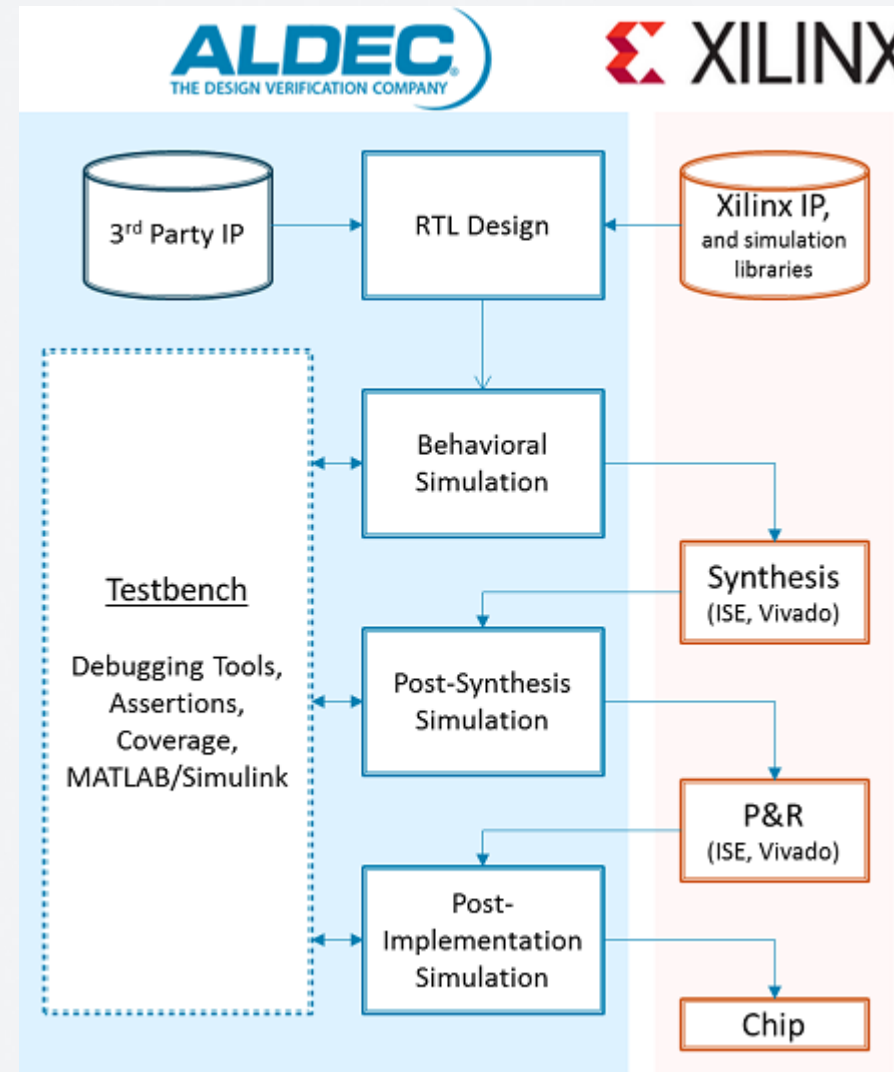
## Introduction to FPGA

- **FPGA** stands for *Field Programmable Gate Array*.
- A reconfigurable digital hardware device.
- Can be programmed **after manufacturing**.
- Used for parallel and high-speed processing.
- Examples: Xilinx, Intel (Altera), Lattice



# FPGA-flow

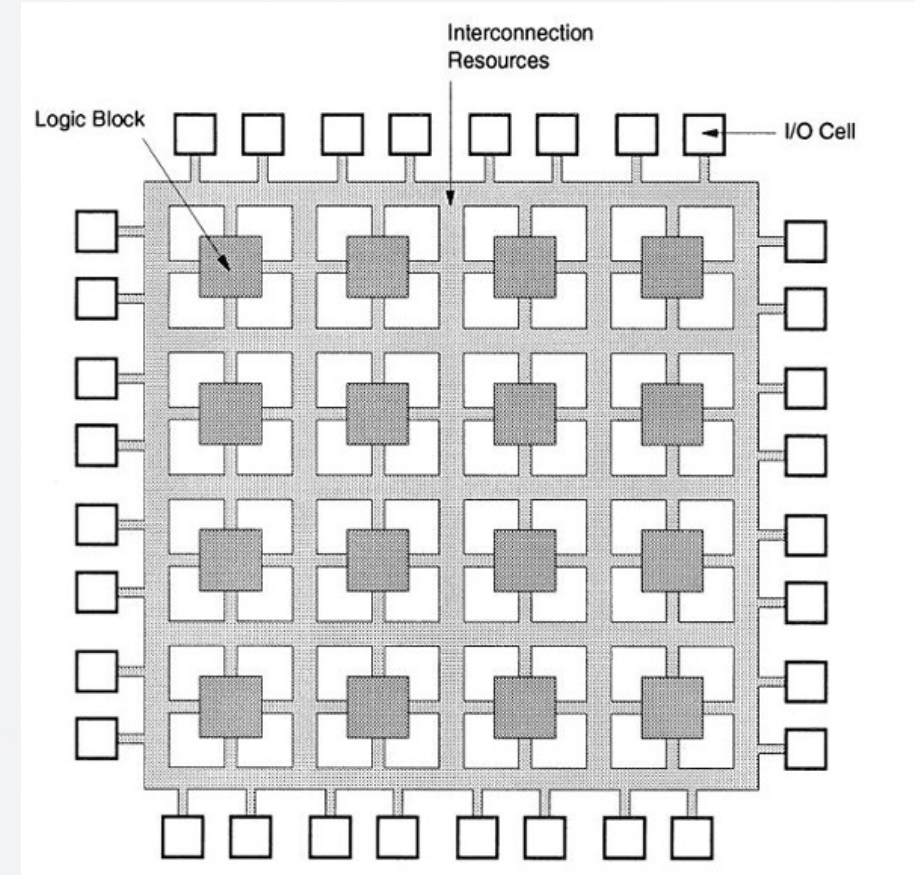
1. Design Entry
2. Functional Simulation
3. Synthesis
4. Implementation
5. Timing Verification
6. Bitstream Generation
7. Programming & Testing



# FPGA-fabric

## FPGA Fabric

- Internal architecture of an FPGA.
- Made of configurable logic blocks.
- Interconnected using programmable routing.
- Enables custom digital hardware design.





# FPGA-fabric

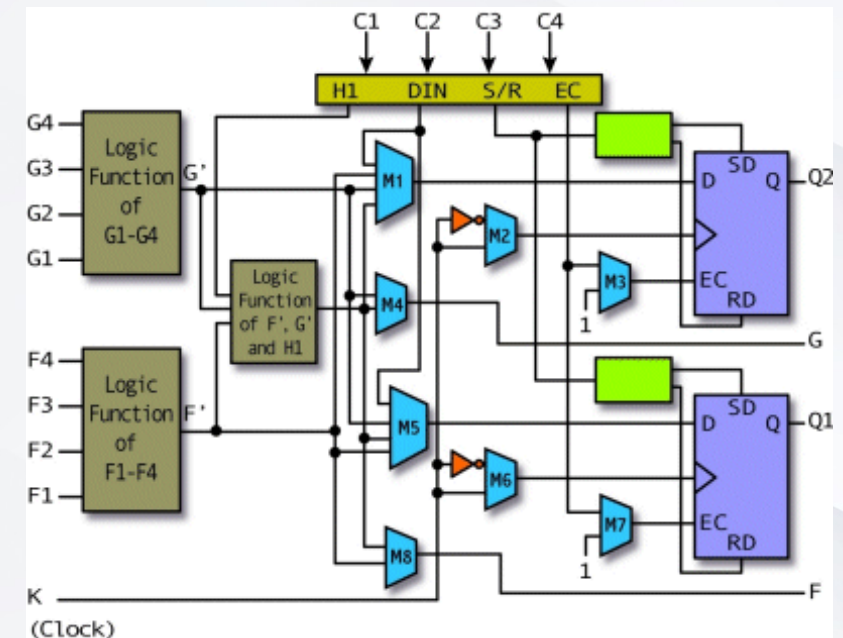
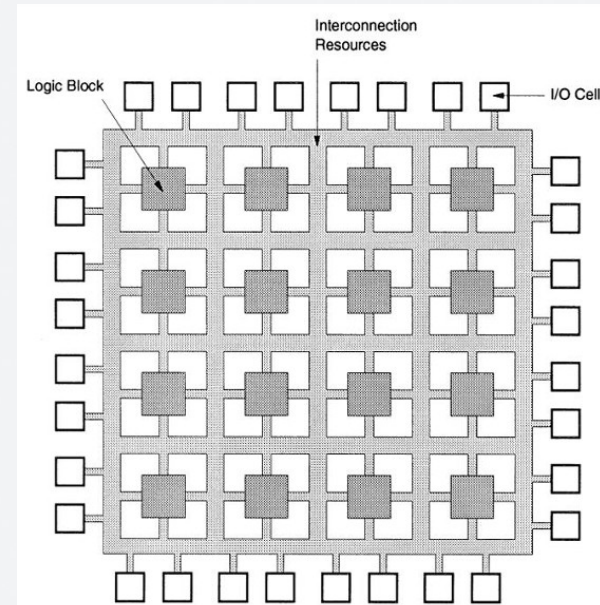
**CLBs (Xilinx): Configurable Logic Blocks**

**LABs (Intel): Logic Array Blocks**

- Basic building blocks of FPGA fabric
- Contain:
  - LUTs (Look-Up Tables)
  - Flip-Flops
- Implement combinational and sequential logic

**Slice (Xilinx) / LE (Intel)**

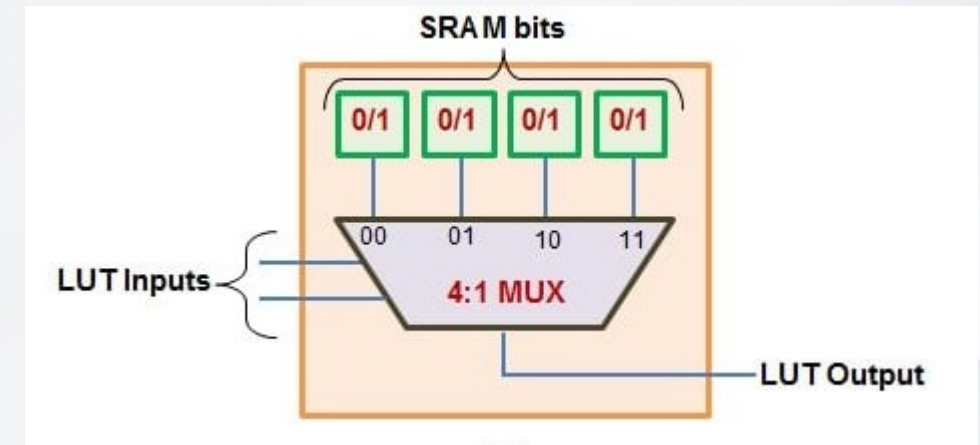
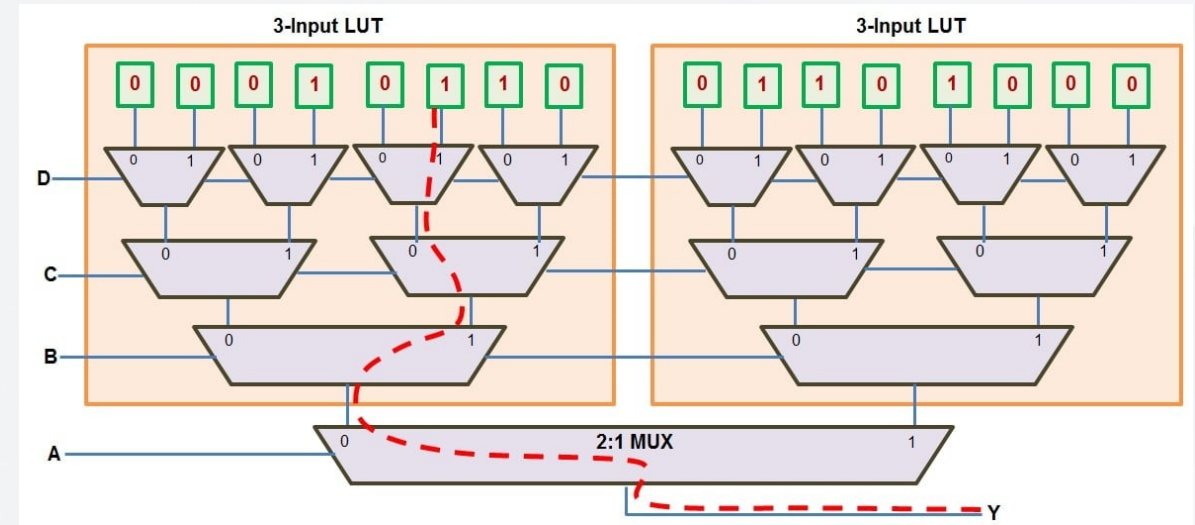
- Subdivision of a CLB
- Contains LUTs and registers



# FPGA-fabric

## Look-Up Tables (LUTs)

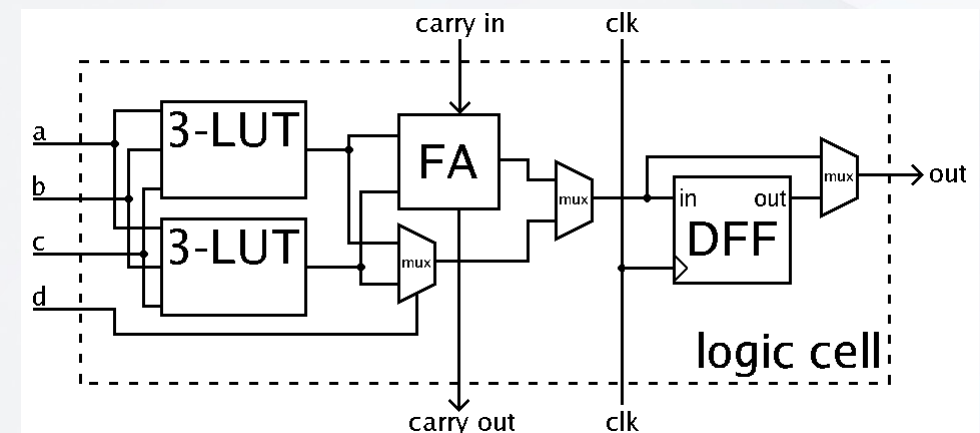
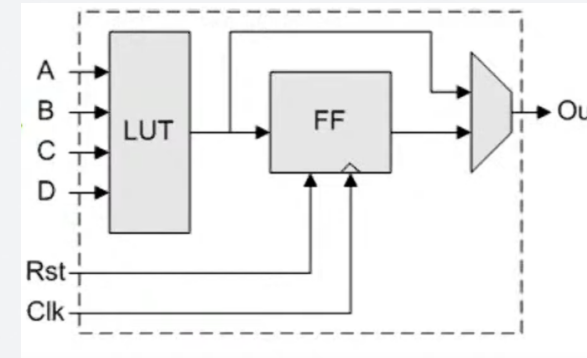
- Implement combinational logic
- Act as small memory blocks
- Outputs precomputed logic results
- Size defined by number of inputs (e.g., 4-LUT, 6-LUT)



# FPGA-fabric

## Flip-Flops

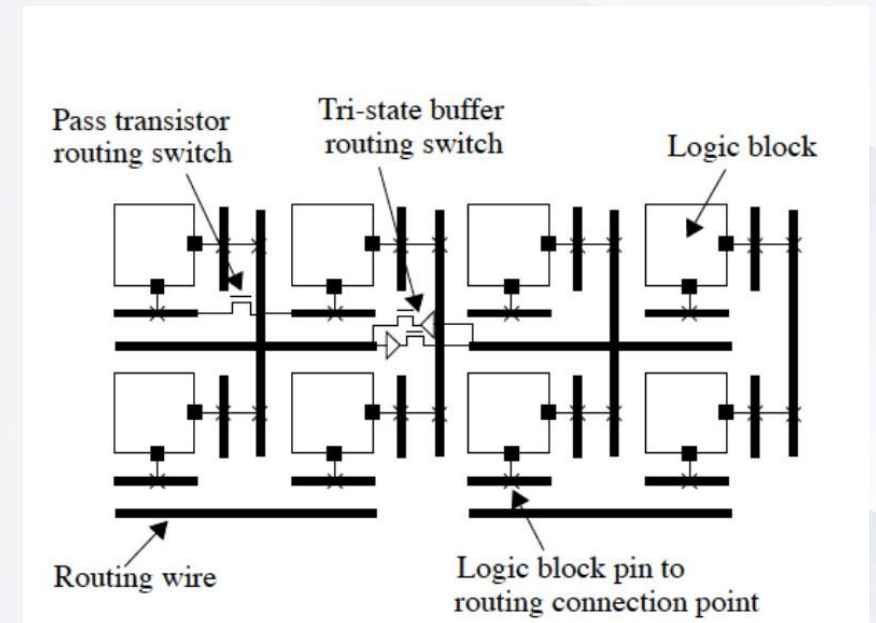
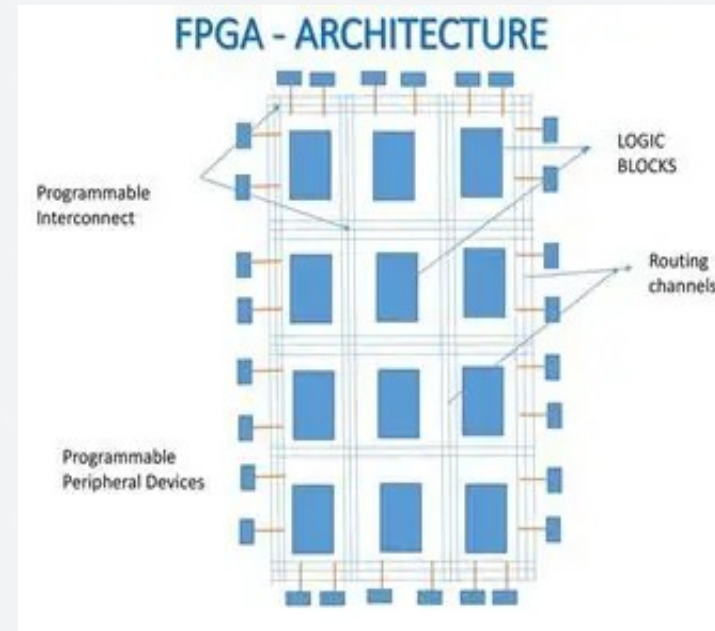
- Store 1 bit of data
- Used for sequential logic
- Clock-driven elements
- Enable pipelining and state machines



# FPGA-fabric

## Programmable Interconnect

- Connects logic blocks together
- Configurable routing paths
- Enables signal communication across FPGA
- Major factor in timing performance

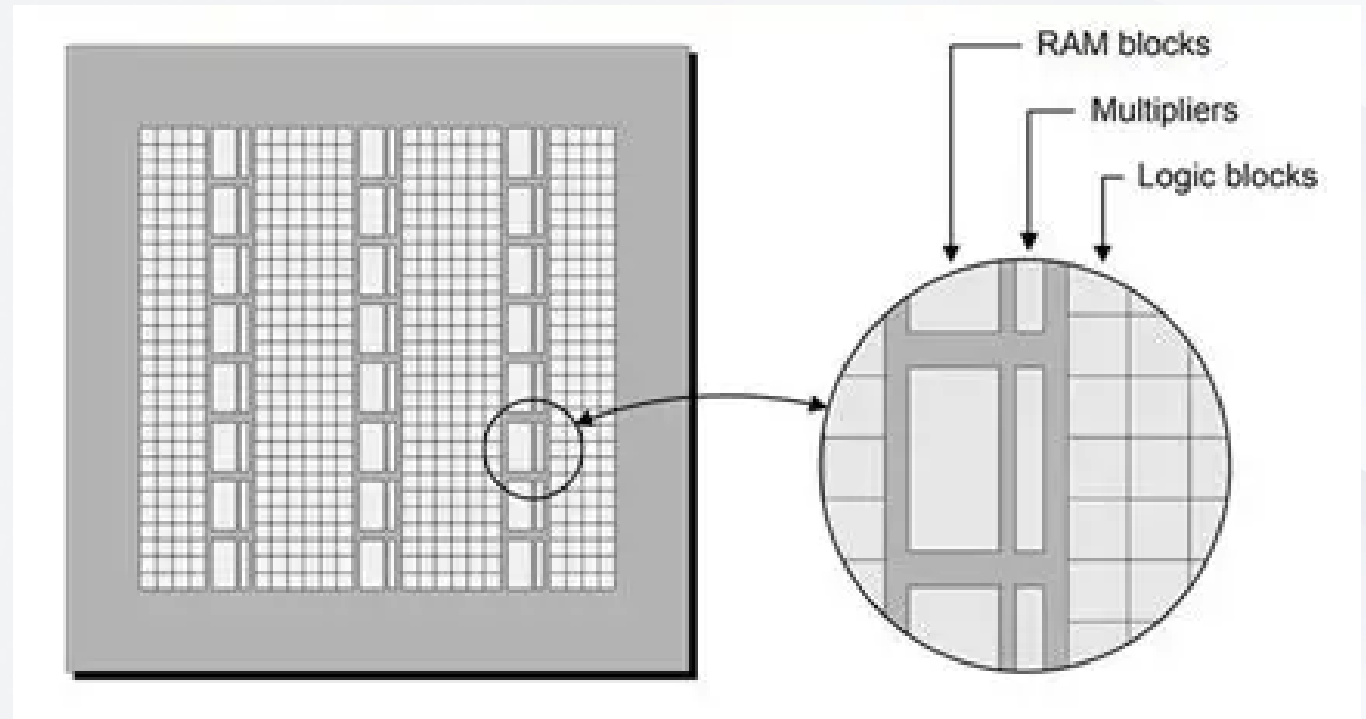




# FPGA-fabric

## Block RAM (BRAM)

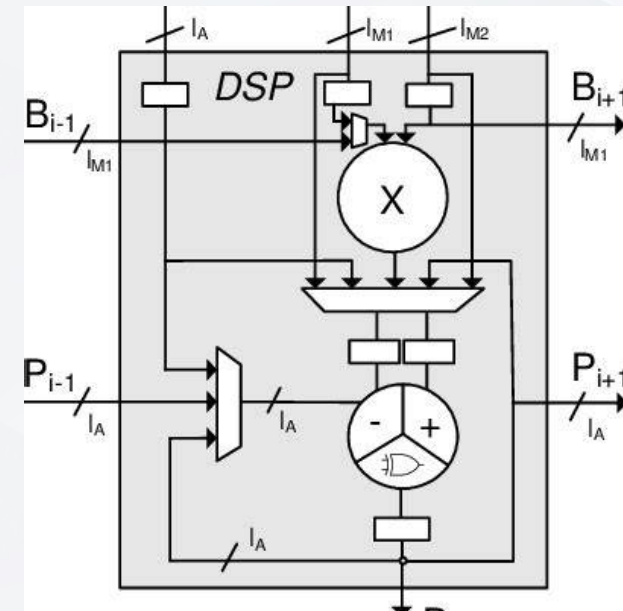
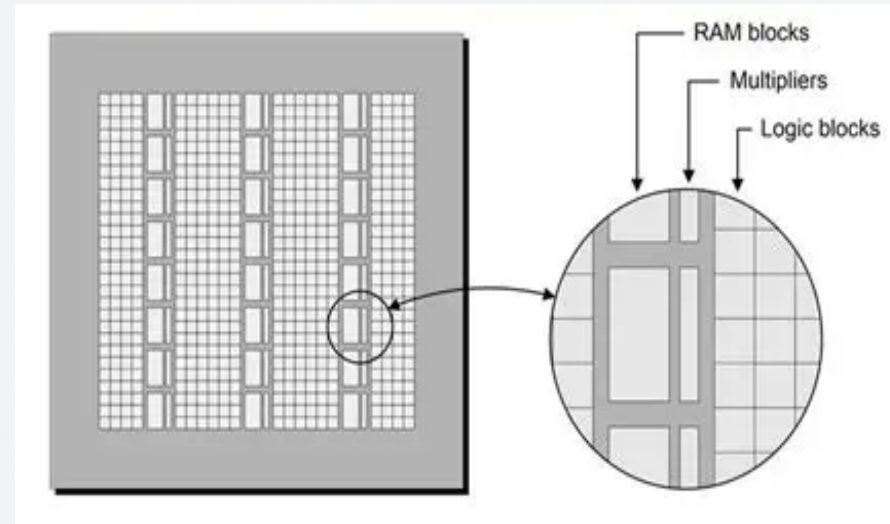
- Dedicated memory blocks
- Used for:
  - Buffers
  - FIFOs
  - Caches
- Faster than LUT-based memory



# FPGA-fabric

## DSP Blocks

- Optimized for arithmetic operations
- Support:
  - Multiplication
  - Addition
  - MAC operations
- Used in signal processing & AI



# Design Styles – Pros and Cons

- **Full Custom Design:**

1. Customization for optimized power, performance, area.
2. High complexity = cost, time-to-market, high risk.

- **Standard Cell:**

1. Simple, fast, reliable.
2. Only Digital designs. Excess power, wirelength, etc.

- **FPGA:**

1. Post silicon configurability, very inexpensive.
2. High percentage of overhead. High cost per chip.

- **Microcontroller (Software)**

1. Programmable, very inexpensive
2. Very slow compared to hardware implementation.

# Exercise 1

- **What is the best design approach for implementing each of the following:  
[hint: you can use a mix of approaches when needed]**
  1. **ADC**
  2. **Microcontroller**
  3. **AI accelerator HW testing and prototyping.**
  4. **Mobile application chip.**
  5. **Medical and Aerospace applications. [Low volume and high complexity]**



# Exercise 2

- **What makes an FPGA relatively consumes more power than a corresponding ASIC?**
- **What makes an FPGA relatively has less performance ( $F_{max}$ ) than a corresponding ASIC?[Please list 3 reasons at least]**
- **When can we consider the high NRE cost of ASIC development make sense economically?**

# “Three Letter Words”

- **PDK – Process Design Kit**
- **HDL – Hardware Description Language**
- **RTL – Register Transfer Level**
- **EDA – Electronic Design Automation**
- **ASIC – Application Specific Integrated Circuit**
- **FPGA – Field Programmable Gate Array**
- **IP – Intellectual Property**
- **DRC – Design Rule Check**
- **LVS – Layout Vs. Schematics**
- **RCX – Resistance/Capacitance Extraction**
- **DFM – Design for Manufacturing**
- **DFT – Design for Testability**
- **UPF - Unified Power Format**
- **PnR – Placement and Routing**

# Thank You!